

Parameter-Free Universal Trust-Region Methods for Nonconvex–Strongly-Concave Minimax Optimization

1 Introduction

We consider smooth nonconvex–strongly-concave (NC–SC) minimax optimization

$$\min_{x \in \mathbb{R}^{d_x}} \max_{y \in \mathbb{R}^{d_y}} f(x, y), \quad (1.1)$$

where $f(x, \cdot)$ is strongly concave for every x , while $f(\cdot, y)$ may be nonconvex. Strong concavity makes the inner maximizer $y^*(x) := \arg \max_y f(x, y)$ well defined and gives rise to the primal function $P(x) := \max_y f(x, y)$. Under standard smoothness assumptions, P is twice differentiable but generally nonconvex. An approximate first-order stationary point of P therefore need not capture local optimality, which motivates the search for approximate second-order stationary points.

Second-order algorithms for NC–SC minimax optimization have been developed using cubic regularization, trust-region methods, and other Newton-type updates [1, 2, 3, 4, 5]. A representative method is Minimax Cubic Newton (MCN) [1], which approximates the gradient and Hessian of P through inner maximization and achieves the characteristic $O(\epsilon^{-3/2})$ second-order oracle complexity. Several subsequent methods attain the same $\epsilon^{-3/2}$ -type outer dependence under related second-order stationarity criteria. More recently, joint-minimization reformulations have led to methods that avoid repeatedly solving the inner maximization [6, 7].

For methods that construct approximate derivatives of P through inner maximization, the outer iteration complexity alone does not describe the total computational work. At each outer iterate x_k , one must approximate $y^*(x_k)$ accurately enough that the resulting first- and second-order information of P can be used reliably. In MCN, accelerated gradient ascent is used for this purpose. Its first-order oracle complexity is stated in $\tilde{O}(\epsilon^{-3/2})$ form, and the explicit bound contains a logarithmic dependence on the target accuracy multiplying the leading outer-order term.

Our method builds on the universal trust-region (UTR) framework of Jiang et al. [8]. We first formulate a general inexact UTR interface in which working queries provide approximate values, gradients, and Hessians, and validation queries check possible solutions. Its convergence theorem controls both the outer iteration count and the cumulative path needed for inner-work analysis. The minimax algorithm realizes this interface through corrected primal-function oracles and persistent calls to SCAR [9]. It retains the inner solver’s curvature estimates across successive maximization problems and adjusts inner accuracy to the current outer scale. This gives the desired leading outer and inner complexity without supplying smoothness or strong-concavity parameters to either component.

1.1 Main Contributions

Our contributions concern the total inner work and the removal of problem parameters from the executable minimax method. The existing implicit gradient and Schur-complement Hessian are ingredients of the construction.

Sharper total inner-gradient complexity. Let ℓ and ρ denote the Lipschitz constants of the joint gradient and Hessian of f , respectively, let $\mu > 0$ be the strong-concavity modulus of $f(x, \cdot)$, and set

$$\kappa := \ell/\mu, \quad \Delta_P := P(x_0) - \inf_x P(x), \quad M_P := 4\sqrt{2}\kappa^3\rho.$$

[Algorithm 2](#) combines the implicit gradient and Schur-complement Hessian with an adaptive UTR step and accesses inner maximization through observable gradient residuals. The key estimate is a cumulative cubic bound on the accepted outer steps, which amortizes the logarithmic costs of successive warm-started inner calls. Persistent SCAR also shares its curvature calibration across the whole run. For each fixed regularity class and sufficiently small ϵ , [Theorem 3.1](#) gives

$$\begin{aligned} Q &= O\left(\Delta_P \kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2}\right) + O_{\text{fixed}}(\log^3(e/\epsilon)), \\ N_y &= O\left(\Delta_P \kappa^2 \sqrt{\rho} \epsilon^{-3/2}\right) + O_{\text{fixed}}(\sqrt{\kappa} \log^3(e/\epsilon)), \end{aligned}$$

where Q counts trust-region subproblems and N_y counts all inner-gradient evaluations. Here O_{fixed} permits constants depending on the fixed payoff, regularity constants, and initialization, but not on ϵ . Neither $\log(1/\epsilon)$ nor $\log \kappa$ multiplies the displayed leading inner term. In particular, this removes the leading target-accuracy logarithm in the explicit inner-gradient bound of MCN. As a technical tool, [Theorem 2.1](#) gives a general inexact UTR complexity bound, supplemented by the path estimates in [Proposition A.1](#); the minimax analysis invokes these results after verifying the corrected-oracle error bounds.

A fully parameter-free second-order method. [Algorithm 2](#) uses only the initialization, target tolerance, and universal numerical constants. Neither the outer scheme nor SCAR requires ℓ , μ , ρ , κ , or M_P as an input. Working inner accuracy is restored through a number of residual halvings determined by the observed outer step and current scale; more accurate validation is reserved for candidate solutions. The output satisfies

$$\|\nabla P(x_{\text{out}})\| < \epsilon, \quad \lambda_{\min}(\nabla^2 P(x_{\text{out}})) \geq -\frac{85}{16\sqrt{6}}\sqrt{M_P\epsilon}.$$

A universal constant rescaling of the internal tolerance gives the standard ϵ -second-order guarantee defined in [Definition 2.2](#). The guarantee is class-uniform asymptotic: after fixing common (ℓ, μ, ρ) , one threshold $\epsilon_0(\ell, \mu, \rho) > 0$ works for every admissible payoff and initialization. The algorithm does not know this threshold or certify that the requested accuracy lies below it. [Section 5](#) explains this qualification through the residual-only obstruction and the role of validation.

[Table 1](#) compares methods whose stated second-order guarantees and oracle counts match the quantities considered here. Cubic-LocalMinimax [2], whose complexity is stated using a different local-minimax stationarity measure, is discussed below without reparameterizing it for the table. The recent single-loop regularized Newton method of Ma et al. [7] has no inner maximization loop of the form measured by the table’s third column.

1.2 Related Work

Implicit differentiation and primal-function estimators. Ablin et al. [10] analyze the implicit gradient estimator for functions defined through optimization and establish its quadratic error in Proposition 2.3. In minimax optimization, Zhang et al. [11] use the corresponding total gradient and Schur-complement Hessian; see their equations (1.1)–(1.2). These estimators are ingredients of the present work. Our contributions concern their integration into an inexact UTR method, the accumulated inner-gradient complexity, and parameter-free tracking across changing inner objectives.

Table 1: Complexity comparison of deterministic second-order methods for smooth NC–SC minimax optimization. Initial optimality gaps and universal numerical constants are suppressed. For cited methods, regularity-parameter dependence is shown when reported in a directly comparable form; entries without such factors report only dependence on ϵ . For our method the entries are leading terms, valid in the class-uniform small-accuracy regime of [Theorem 3.1](#).

Method	Outer complexity	Total inner-gradient evaluations	Parameter-free
MCN [1]	$O(\kappa^{3/2}\sqrt{\rho}\epsilon^{-3/2})$	$\tilde{O}(\kappa^2\sqrt{\rho}\epsilon^{-3/2})$	No
MINIMAX-TR / MINIMAX-TRACE [3]	$O(\epsilon^{-3/2})$	Not reported	No
GRTR / LMNegCur [4]	$\tilde{O}(\kappa^{3/2}\sqrt{\rho}\epsilon^{-3/2})$	$\tilde{O}(\kappa^2\sqrt{\rho}\epsilon^{-3/2})$	No
HSDA / IHSDA [5]	$\tilde{O}(\epsilon^{-3/2})$	Not reported	No
Algorithm 2 (ours)	$O(\kappa^{3/2}\sqrt{\rho}\epsilon^{-3/2})$	$O(\kappa^2\sqrt{\rho}\epsilon^{-3/2})$	Yes

Here $\kappa = \ell/\mu$ and ρ is the Hessian-Lipschitz constant of f . The notation $\tilde{O}(\cdot)$ is retained when used in the cited reference. *Not reported* means that the cited work does not state a separate comparable total inner-gradient bound. The complete bounds for [Algorithm 2](#) include additive $O_{\text{fixed}}(\log^3(e/\epsilon))$ and $O_{\text{fixed}}(\sqrt{\kappa} \log^3(e/\epsilon))$ terms for the outer and inner counts, respectively. Its executable steps use no problem parameters; the accuracy threshold in its guarantee is unknown.

First-order methods for NC–SC minimax optimization. First-order methods for NC–SC minimax problems have been studied extensively. Representative approaches include gradient descent-ascent, multi-step ascent, proximal schemes, and accelerated methods [12, 13, 14]. Their complexity for finding first-order stationary points of the primal function has also been studied from both upper- and lower-bound perspectives; see, for example, Zhang et al. [15] and the references therein. These results form the first-order counterpart to the second-order stationarity problem considered in this paper.

Second-order methods for NC–SC minimax optimization. Luo et al. [1] proposed MCN, which alternates accelerated gradient ascent for approximating $y^*(x)$ with a cubic-regularized update based on approximate first- and second-order information of the primal function. Its leading bounds are $O(\Delta_P \kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2})$ second-order oracle evaluations and $\tilde{O}(\Delta_P \kappa^2 \sqrt{\rho} \epsilon^{-3/2})$ first-order oracle evaluations. Chen et al. [2] proposed Cubic-LocalMinimax and developed inexact and stochastic variants, using cubic regularization to obtain global convergence toward local minimax points. Their complexity is formulated using a different stationarity measure, so we do not convert it to the accuracy parameterization used in [Table 1](#).

Trust-region approaches were developed by Yao and Xu [3], who proposed MINIMAX-TR and MINIMAX-TRACE. Wang and Xu [4] subsequently proposed gradient-norm regularized trust-region and Levenberg–Marquardt methods, together with inexact variants, and established $\tilde{O}(\kappa^{3/2}\sqrt{\rho}\epsilon^{-3/2})$ outer complexity and $\tilde{O}(\kappa^2\sqrt{\rho}\epsilon^{-3/2})$ total gradient-ascent complexity, suppressing the initial-gap dependence. More recently, Chen et al. [5] proposed homogeneous second-order descent-ascent methods and developed an inexact Lanczos implementation for large-scale problems.

A recent alternative is to avoid repeated inner maximization altogether. Ma et al. [6] introduced an equivalent joint-minimization reformulation that preserves first- and second-order stationarity and used it to develop a parameter-free line-search framework. Building on this reformulation, Ma et al. [7] proposed a single-loop regularized Newton method with $O(\epsilon^{-3/2})$ global iteration complexity and local superlinear convergence. This line of work addresses the cost of the inner loop by changing the formulation. Its analyses

use the regularized joint objective

$$h_\beta(x, y) = f(x, y) + \frac{\beta}{2} \|\nabla_y f(x, y)\|^2.$$

In particular, the deterministic single-loop bound depends on $h_\beta(x_0, y_0) - \inf_x P(x)$, rather than only on the primal gap $P(x_0) - \inf_x P(x)$, and measures finite-accuracy stationarity of h_β [7, Theorem 3.9]. The dependence on the joint initialization and regularization parameter should therefore be retained when comparing these bounds. The present paper instead studies total inner-gradient complexity when approximate derivatives of P are constructed through inner maximization.

Adaptive and parameter-free optimization. Adaptive optimization methods aim to reduce or eliminate prior knowledge of problem regularity constants. In smooth nonconvex optimization, the universal trust-region framework [8] uses gradient-scaled trust-region updates and provides the outer mechanism used here. On the inner side, Lan et al. [9] developed SCAR, a parameter-free accelerated method for smooth strongly convex optimization that adapts to unknown smoothness and strong-convexity parameters. The general inexact UTR theorem and the SCAR guarantees are presented as tools in Section 2; their supporting proofs are deferred to the appendices.

Related adaptive and parameter-free methods have also been developed for minimax optimization and monotone variational inequalities [16, 17]. More recently, Wang et al. [18] proposed a fully parameter-free second-order method for convex–concave minimax optimization, requiring neither the Hessian-Lipschitz constant nor an a priori distance bound to a solution. For NC–SC minimax problems, Ma et al. [6] developed a parameter-free line-search framework based on a joint-minimization reformulation. Our method retains the primal-function formulation and combines a gradient-scaled UTR step with persistent SCAR tracking across successive inner maximization problems.

1.3 Organization

Section 2 introduces the inexact UTR and SCAR tools and the NC–SC setting. Section 3 presents the corrected oracles, their SCAR realization, and the minimax complexity theorem. Section 4 proves that the minimax oracles satisfy the UTR interface and bounds their total inner work. Section 5 discusses the scope of the guarantee, and Section 7 concludes the paper.

2 Notations and preliminaries

The norm $\|\cdot\|$ is Euclidean for vectors and spectral for matrices. For a symmetric matrix H , $\lambda_{\min}(H)$ denotes its smallest eigenvalue. Unsubscripted logarithms are natural. Universal $O(\cdot)$ constants do not depend on problem parameters. For $b_\epsilon \geq 0$, the relation $a_\epsilon = O_{\text{fixed}}(b_\epsilon)$ as $\epsilon \downarrow 0$ means that, for every fixed choice of the payoff, regularity constants, and initialization, there exist $C > 0$ and $\epsilon' > 0$ such that $|a_\epsilon| \leq Cb_\epsilon$ for all $0 < \epsilon \leq \epsilon'$. These constants need not be uniform over problem instances. The notation $\tilde{O}(\cdot)$ suppresses logarithmic factors.

Definition 2.1 (Lipschitz continuity). *A map G between normed spaces is L -Lipschitz on a set D if*

$$\|G(u) - G(v)\| \leq L\|u - v\| \quad (u, v \in D).$$

For gradients we use the Euclidean norm, and for Hessians the spectral norm. For a joint variable (x, y) , the domain norm is $\|(x, y)\| = (\|x\|^2 + \|y\|^2)^{1/2}$.

Definition 2.2 (Approximate second-order stationarity). *Let F be twice continuously differentiable with an M -Lipschitz Hessian, where $M > 0$. For $\epsilon > 0$, a point x is an ϵ -second-order stationary point (ϵ -SOSP) if*

$$\|\nabla F(x)\| \leq \epsilon, \quad \lambda_{\min}(\nabla^2 F(x)) \geq -\sqrt{M\epsilon}.$$

Lemma 2.1 (Trust-region optimality conditions). *Let B be symmetric, $g \in \mathbb{R}^n$, and $R > 0$. A vector d is a global minimizer of $g^T d + d^T B d / 2$ over $\|d\| \leq R$ if and only if there exists $\lambda \geq 0$ such that*

$$(B + \lambda I)d = -g, \quad B + \lambda I \succeq 0, \quad \|d\| \leq R, \quad \lambda(\|d\| - R) = 0. \quad (2.1)$$

This characterization is standard; see Luenberger and Ye [19, Section 8.7].

All trust-region subproblems below are solved globally, and the solver returns a pair (d, λ) satisfying (2.1).

2.1 Parameter-free UTR with inexact oracles

We first isolate an outer method for minimizing a general objective F . It uses approximate values, gradients, and Hessians through a working oracle $\mathcal{W}$, and checks possible solutions through a validation oracle $\mathcal{V}$. The oracle calls may use auxiliary data, such as an inner iterate, and retain solver state between calls. Their accuracy requirements below are analytical assumptions, not error certificates that the algorithm is required to compute.

Write $w = (\hat{F}, \hat{g}, \hat{H}; \xi)$ for the stored working data: the first three entries approximate the value, gradient, and Hessian, and ξ contains auxiliary data such as an inner iterate. We distinguish three working operations:

$$\begin{aligned} w_0 &= \mathcal{W}_{\sigma_0}^0(x_0), & \text{initialization,} \\ w^+ &= \mathcal{W}_{\sigma, x}(x^+; w), & \text{trial at } x^+ \text{ from data stored at } x, \\ w_{\text{new}} &= \mathcal{W}_{2\sigma}^r(x; w), & \text{refresh at } x \text{ after doubling the scale.} \end{aligned} \quad (2.2)$$

The validation call is

$$(g_v, H_v) = \mathcal{V}_\sigma(x^+; w^+). \quad (2.3)$$

Here $\mathcal{W}$ collectively denotes its initialization, trial, and refresh operations. Unadorned hats denote the components of w , and superscript $+$ denotes components of w^+ . These are stateful calls, not necessarily fixed functions: shared solver calibration may change on any call, whereas the supplied point data w are unchanged unless explicitly reassigned. This distinction also separates a zero-length trial from a refresh.

Assumption 2.1 (Inexact working and validation oracles). *Fix $\epsilon > 0$ and $\sigma_0 > 0$. All calls in (2.2)–(2.3) terminate in finite time and return symmetric Hessian estimates. For constants $C_g, C_v \geq 0$, the numerical components of every working output at a query point x and generated scale $\sigma \geq \sigma_0$ satisfy*

$$\|\hat{g} - \nabla F(x)\| \leq C_g \frac{\epsilon}{\sigma^2}, \quad |\hat{F} - F(x)| \leq C_v \frac{\epsilon^{3/2}}{\sigma^3}. \quad (2.4)$$

Whenever $\sigma^2 \geq M/6$, it additionally satisfies

$$\|\hat{g} - \nabla F(x)\| \leq \frac{\epsilon}{1024}, \quad \|\hat{H} - \nabla^2 F(x)\| \leq \frac{\sigma\sqrt{\epsilon}}{64}, \quad |\hat{F} - F(x)| \leq \frac{\epsilon^{3/2}}{4096\sigma}. \quad (2.5)$$

Every output $(g_v, H_v) = \mathcal{V}_\sigma(x; w)$, including those with $\sigma^2 < M/6$, satisfies

$$\|g_v - \nabla F(x)\| \leq \frac{\epsilon}{1024}, \quad \|H_v - \nabla^2 F(x)\| \leq \frac{\sigma\sqrt{\epsilon}}{32}. \quad (2.6)$$

The requirements apply to every actual call, including repeated queries and refreshes. They are analytical accuracy conditions; M, C_g, C_v are not algorithm inputs. Exact information is a special case: each working operation returns $(F(x), \nabla F(x), \nabla^2 F(x); \emptyset)$, independently of its scale and warm start, and validation returns $(\nabla F(x), \nabla^2 F(x))$. All errors are then zero, so the assumption holds with $C_g = C_v = 0$ for every $\epsilon, \sigma_0 > 0$; no refinement or auxiliary data are needed.

At a stored working tuple, set

$$h := \max\{\|\hat{g}\|, \epsilon\}, \quad R(h, \sigma) := \frac{\sqrt{h}}{4\sigma}, \quad (2.7)$$

and solve

$$\min_{\|d\| \leq R(h, \sigma)} \hat{g}^T d + \frac{1}{2} d^T \left(\hat{H} + \frac{\sigma\sqrt{\epsilon}}{64} I + \sigma\sqrt{h} I \right) d. \quad (2.8)$$

For its global solution and multiplier (d, λ) , define

$$r := \|d\|, \quad W := \sigma\sqrt{h} r^2, \quad U := \lambda r^2, \quad T := \frac{\epsilon^{3/2}}{\sigma}. \quad (2.9)$$

An ordinary trial is accepted when

$$\hat{F} - \hat{F}^+ \geq \frac{W}{8} + \frac{U}{4} - \frac{T}{1024}, \quad \|\hat{g}^+\| \leq \frac{h}{2} + \lambda r. \quad (2.10)$$

The exception is an interior step at $h = \epsilon$, which invokes validation rather than the ordinary acceptance test. We denote by $\text{TR}_\sigma(w)$ a globally optimal primal–multiplier pair for (2.8), with h defined by (2.7).

Remark 2.1 (Why working and validation oracles?). *The two roles separate the information needed to make progress from the information needed to return a solution. Working tuples build the model (2.8) and evaluate the acceptance tests (2.10). Their accuracy adapts to the current scale; the small-error bounds (2.5) are required only once $\sigma^2 \geq M/6$. Since M is unknown, an interior step at $h = \epsilon$ merely nominates a candidate: at an underestimated scale, the working derivatives need not justify termination.*

Validation checks that candidate with derivative-error bounds (2.6) that hold at every generated scale, including $\sigma^2 < M/6$. It needs no function value because it is used only to test stationarity. Thus the ordinary trials use working information, whereas only potential stopping points receive a validation query. The two oracles need not be different solvers: in the minimax construction, both use corrected estimates computed with SCAR, and validation refines the candidate’s inner iterate. The stated error bounds are proved under the theorem’s assumptions, not computed as certificates by the algorithm.

For a concrete example, in the minimax implementation $\mathcal{W}_{\sigma,z}(x;w)$ starts from the inner point stored at z and performs enough residual halvings to track the move to x . The call $\mathcal{V}_\sigma(x;w^+)$ continues from that candidate inner point to a tighter residual target; see Algorithm 2. With exact derivatives, by contrast, both roles can simply reuse the same exact information.

Related safeguards occur in established model-based methods. Dynamic-accuracy adaptive regularization refines derivative approximations for its optimality test separately from the accuracy requirements for constructing a step [20, Algorithms 3.1–3.3]. Derivative-free trust-region methods use a criticality step to improve model accuracy when the model’s stationarity measure is small [21, Algorithms 4.1–4.2 and 6.1–6.2]. These are conceptual analogies, not instances of our precise two-oracle interface: a small model gradient prompts an accuracy check rather than immediate termination.

Algorithm 1: Parameter-free inexact adaptive UTR

Input: $x_0, \epsilon > 0, \sigma_0 > 0, \mathcal{W}, \mathcal{V}$
1 $(x, \sigma) \leftarrow (x_0, \sigma_0); w \leftarrow \mathcal{W}_\sigma^0(x)$
2 **while** *true* **do**
3 $h \leftarrow \max\{\|\widehat{g}\|, \epsilon\}; (d, \lambda) \leftarrow \text{TR}_\sigma(w)$
4 $(x^+, r) \leftarrow (x + d, \|d\|); (W, U, T) \leftarrow (\sigma\sqrt{h}r^2, \lambda r^2, \epsilon^{3/2}/\sigma)$
5 $w^+ \leftarrow \mathcal{W}_{\sigma, x}(x^+; w)$
6 **if** $h = \epsilon$ **and** $r < \sqrt{h}/(4\sigma)$ **then**
7 $(g_v, H_v) \leftarrow \mathcal{V}_\sigma(x^+; w^+)$
8 **if** $\|g_v\| \leq \epsilon/2$ **and** $\lambda_{\min}(H_v) \geq -(21/8)\sigma\sqrt{\epsilon}$ **then**
9 **return** x^+
10 **end**
11 **else**
12 **if** (2.10) *holds* **then**
13 $(x, w) \leftarrow (x^+, w^+)$
14 **continue**
15 **end**
16 **end**
17 $\sigma \leftarrow 2\sigma$
18 $w \leftarrow \mathcal{W}_\sigma^r(x; w)$
19 **end**

In Algorithm 1, line 5 forms w^+ without changing w . Apart from initialization, w changes only on acceptance (line 13) or refresh (line 18); there is no unrecorded refinement of a stored value. Successful validation exits at line 9. Otherwise, a trial that does not reach line 13 doubles σ at line 17 and refreshes from the old w . Neither w^+ nor the validation data replace w on this branch, although shared solver calibration is retained. The nondecreasing scale and these explicit assignments allow the proof to sum changes of the inexact function-value proxy. The outer method uses no regularity or error constants. A fully parameter-free implementation also requires parameter-free oracle routines; Section 3 supplies them for minimax optimization.

Theorem 2.1 (Inexact UTR complexity). *Let $F : \mathbb{R}^n \rightarrow \mathbb{R}$ be bounded below and twice continuously differentiable with an M -Lipschitz Hessian, $M > 0$. For $\epsilon, \sigma_0 > 0$, suppose the working and validation oracles satisfy Assumption 2.1 with constants C_g, C_v . Then Algorithm 1, with globally solved subproblems, terminates at a point satisfying*

$$\|\nabla F(x_{\text{out}})\| < \epsilon, \quad \lambda_{\min}(\nabla^2 F(x_{\text{out}})) \geq -\frac{85}{32} \max\{\sigma_0, 2\sqrt{M/6}\}\sqrt{\epsilon}. \quad (2.11)$$

Its total number of trust-region subproblem solves is

$$Q = O\left(\max\{\sigma_0, \sqrt{M}\} \left[\frac{F(x_0) - \inf F}{\epsilon^{3/2}} + \frac{C_v}{\sigma_0^3} + \frac{1}{\sigma_0} \log\left(e + \frac{\|\nabla F(x_0)\|}{\epsilon} + \frac{C_g}{\sigma_0^2}\right) \right] + \log\left(1 + \frac{\sqrt{M}}{\sigma_0}\right)\right), \quad (2.12)$$

with a universal hidden constant. The algorithm does not require M, C_g , or C_v .

The theorem separates outer progress from the construction and cost of the oracles. Its proof is in Section A. The finer counting and path estimates used in the minimax analysis are recorded separately in Proposition A.1; they do not require any additional assumption or algorithmic input.

2.2 SCAR and persistent residual reduction

The inner problems use the parameter-free SCAR method of Lan et al. [9]. We specify its accumulative regularization (AR) routine and two residual-reduction interfaces, importing only the guarantee needed for persistent use across changing objectives. All oracle counts below count gradient evaluations; line searches also use function values.

The AR building block. The procedure below is Lan et al. [9, Algorithm 4.2], with their backtracking routine (Algorithm 4.1) written inline. We use α_s for its regularization to distinguish it from the outer UTR scale σ . Its accelerated subsolver $\mathcal{A}$ is the line-search implementation described after their Assumption 4.1, treating the quadratic regularizer as a proximable term. For a convex objective ϕ with an L -Lipschitz gradient, define

$$\phi_s(w) := \phi(w) + \frac{\alpha_s}{2} \|w - \bar{u}_s\|^2, \quad u_s^* := \arg \min_w \phi_s(w),$$

The subsolver, started at u_{s-1} , supplies iterates u_s^k and computed estimates L_s^k satisfying

$$\phi_s(u_s^k) - \phi_s(u_s^*) \leq \frac{L_s^k}{k^2} \|u_{s-1} - u_s^*\|^2, \quad 0 < L_s^k \leq 4L. \quad (2.13)$$

We write this stopped solver as

$$\mathcal{A}(\phi, \alpha_s, \bar{u}_s, u_{s-1}) := u_s^{k_s}, \quad k_s := \min \left\{ k \geq 1 : k \geq 8\sqrt{2L_s^k/\alpha_s} \right\}.$$

The index k_s is determined along the generated trajectory, not specified in advance. Each call starts with fresh accelerated momentum. Only the displayed computable stopping test is used; L and u_s^* occur in the guarantee, not in the input.

Procedure AR($\phi, u, \alpha, \mathcal{M}$)

Input: $\phi : \mathbb{R}^m \rightarrow \mathbb{R}$, $u \in \mathbb{R}^m$, $\alpha, \mathcal{M} > 0$

Output: $(v, \mathcal{M}^+) \in \mathbb{R}^m \times (0, \infty)$

```

1  $u_0 \leftarrow u$ ,  $\bar{u}_0 \leftarrow u$ ,  $\alpha_0 \leftarrow 0$ ,  $\mathcal{M}_0 \leftarrow \mathcal{M}$ 
2 for  $s = 1, 2, \dots$  do
3    $\alpha_s \leftarrow 4^{s-1}\alpha$ ,  $\bar{u}_s \leftarrow (\alpha_{s-1}/\alpha_s)\bar{u}_{s-1} + (1 - \alpha_{s-1}/\alpha_s)u_{s-1}$ 
4    $u_s \leftarrow \mathcal{A}(\phi, \alpha_s, \bar{u}_s, u_{s-1})$ 
5    $\mathcal{M}_s \leftarrow \mathcal{M}_{s-1}/2$ ,  $p_s \leftarrow \nabla \phi_s(u_s)$ 
6    $w_s \leftarrow u_s - p_s/[2(\mathcal{M}_s + \alpha_s)]$ 
7   while  $\phi_s(w_s) - \phi_s(u_s) - p_s^T(w_s - u_s) > \frac{\mathcal{M}_s + \alpha_s}{2} \|w_s - u_s\|^2$  do
8      $\mathcal{M}_s \leftarrow 2\mathcal{M}_s$ ,  $w_s \leftarrow u_s - p_s/[2(\mathcal{M}_s + \alpha_s)]$ 
9   end
10  if  $\alpha_s \geq \mathcal{M}_s$  then
11    return  $(u_s, \mathcal{M}_s)$ 
12  end
13 end
```

In AR, line 4 calls the stopped accelerated solver, and line 8 doubles the trial estimate until the local Taylor inequality holds. The point w_s is used only for that test: line 11 returns u_s , not w_s . The returned $\mathcal{M}^+$ is not a certified global upper bound on L .

Fact 2.1 (AR guarantee; Lan et al. [9, Theorem 4.1 and Algorithm 4.2]). *Let $\phi : \mathbb{R}^m \rightarrow \mathbb{R}$ be differentiable and μ -strongly convex with an L -Lipschitz gradient, where $0 < \mu \leq L$. For any $u \in \mathbb{R}^m$, $\nu > 0$, and $0 < \mathcal{M} \leq 4L$, the call*

$$(v, \mathcal{M}^+) = \text{AR}(\phi, u, \nu/10, \mathcal{M})$$

uses $O(1 + \sqrt{L/\nu})$ gradient evaluations, returns $0 < \mathcal{M}^+ \leq 2L$, and satisfies

$$\text{if } \nu \leq \mu, \quad \|\nabla\phi(v)\| \leq \frac{1}{2}\|\nabla\phi(u)\|. \quad (2.14)$$

The hidden constant is universal, and AR requires neither L nor μ as an input.

The residual-halving implication is the cited distance-to-solution bound combined with strong convexity; the short derivation is given in [Section B](#).

SCAR as repeated residual halving. The next two procedures write Lan et al. [9, Algorithm 5.1] as a halving call followed by a residual-target loop, with an initial residual check. The pair $(\nu, \mathcal{M})$ is passed by reference: every update is retained by the caller, even if an AR candidate fails the halving test.

Procedure SCARHalf($\phi, u; \nu, \mathcal{M}$)	
Input: $\phi : \mathbb{R}^m \rightarrow \mathbb{R}$, $u \in \mathbb{R}^m$, $\nu, \mathcal{M} > 0$	
Output: $v \in \mathbb{R}^m$, $\ \nabla\phi(v)\ \leq \ \nabla\phi(u)\ /2$; $(\nu, \mathcal{M}) \in (0, \infty)^2$	
1	$R \leftarrow \ \nabla\phi(u)\ $
2	if $R = 0$ then
3	return u
4	end
5	while $R > 0$ do
6	$(v, \mathcal{M}) \leftarrow \text{AR}(\phi, u, \nu/10, \mathcal{M})$
7	if $\ \nabla\phi(v)\ \leq R/2$ then
8	return v
9	end
10	$\nu \leftarrow \nu/4$
11	end

Procedure SCARTarget($\phi, u, \tau; \nu, \mathcal{M}$)	
Input: $\phi : \mathbb{R}^m \rightarrow \mathbb{R}$, $u \in \mathbb{R}^m$, $\tau, \nu, \mathcal{M} > 0$	
Output: $u \in \mathbb{R}^m$, $\ \nabla\phi(u)\ \leq \tau$; $(\nu, \mathcal{M}) \in (0, \infty)^2$	
1	while $\ \nabla\phi(u)\ > \tau$ do
2	$u \leftarrow \text{SCARHalf}(\phi, u; \nu, \mathcal{M})$
3	end
4	return u

In [SCARHALF](#), line 1 fixes the initial residual, and line 6 updates the shared $\mathcal{M}$ even when the candidate fails. On failure, line 10 changes only ν : the starting point u and residual R stay fixed throughout the call. By contrast, line 2 of [SCARTARGET](#) replaces its iterate after each successful halving and stops once the prescribed target is met. Both procedures retain every update to $(\nu, \mathcal{M})$.

For a single objective, initializing $(\nu, \mathcal{M})$ once and running SCARTARGET implements SCAR with an initial stopping check. We use the name SCARTARGET for this target-based interface and SCARHALF for one successful halving. Both will be analyzed through [Fact 2.1](#), with calibration retained across calls.

An admissible initialization is obtained from two gradient evaluations. For any $z \neq u$, define

$$a(\phi; u, z) := \frac{\|\nabla\phi(z) - \nabla\phi(u)\|}{\|z - u\|}. \quad (2.15)$$

Strong convexity and gradient Lipschitz continuity imply

$$\mu \leq a(\phi; u, z) \leq L. \quad (2.16)$$

Thus $\nu_0 = \mathcal{M}_0 = a(\phi; u, z)$ initializes SCAR without knowing either problem parameter. For the minimax problem below, this construction is applied to $\phi = -f(x, \cdot)$.

Persistent use across inner problems. Repeatedly initializing SCAR would repeat its curvature calibration. Our implementation instead retains the same $(\nu, \mathcal{M})$ across all calls to SCARHALF and SCAR-TARGET, even when ϕ changes. Inner iterates may be warm-started, but accelerated momentum and the internal AR stages are restarted at each AR call. The following consequence of the AR guarantee justifies this cross-objective reuse. Its proof is in [Section B](#); the sequential guarantee is not attributed to the original SCAR theorem. In particular, a successful halving does not certify $\nu \leq \mu$; the proof uses only the implication that a failed halving requires $\nu > \mu$.

Lemma 2.2 (Persistent SCAR halving). *Let $\{\phi_j\}$ be any sequence of differentiable functions on $\mathbb{R}^m$, all μ -strongly convex and all with L -Lipschitz gradients. Initialize a shared pair $(\nu, \mathcal{M})$ with $\mu \leq \nu_0 \leq L$ and $0 < \mathcal{M}_0 \leq 4L$.*

Every call to SCARHALF terminates and halves its initial residual. Retaining $(\nu, \mathcal{M})$ across any N such calls, including calls on different objectives or from unrelated initial points, gives a total of

$$O\left(\sqrt{\frac{L}{\mu}} \left[N + \left\lceil \log_4 \frac{\nu_0}{\mu} \right\rceil\right]\right) \quad (2.17)$$

gradient evaluations, including the initial residual queries.

2.3 The NC–SC setting

The minimax objective is

$$\min_{x \in \mathbb{R}^{d_x}} P(x), \quad P(x) := \max_{y \in \mathbb{R}^{d_y}} f(x, y). \quad (2.18)$$

We impose the following global regularity assumptions, as in Luo et al. [1].

Assumption 2.2 (NC–SC regularity). *The payoff f is twice continuously differentiable and satisfies the following conditions, with Lipschitz continuity understood as in [Definition 2.1](#):*

- (i) ∇f is ℓ -Lipschitz on $\mathbb{R}^{d_x} \times \mathbb{R}^{d_y}$;
- (ii) $\nabla^2 f$ is ρ -Lipschitz on $\mathbb{R}^{d_x} \times \mathbb{R}^{d_y}$ for some $\rho > 0$;
- (iii) $f(x, \cdot)$ is μ -strongly concave for every x , where $0 < \mu \leq \ell$;
- (iv) $P^* := \inf_x P(x) > -\infty$.

We write $\kappa := \ell/\mu$.

Strong concavity makes $y^*(x) := \arg \max_y f(x, y)$ unique and guarantees that $\nabla_{yy}^2 f(x, y)$ is nonsingular. Define the Schur complement

$$\mathcal{H}(x, y) := \nabla_{xx}^2 f(x, y) - \nabla_{xy}^2 f(x, y) (\nabla_{yy}^2 f(x, y))^{-1} \nabla_{yx}^2 f(x, y). \quad (2.19)$$

Lemma 2.3 (Value-function geometry). *Under [Assumption 2.2](#), P is twice continuously differentiable, and*

$$\nabla P(x) = \nabla_x f(x, y^*(x)), \quad (2.20a)$$

$$\nabla^2 P(x) = \mathcal{H}(x, y^*(x)). \quad (2.20b)$$

Moreover, ∇P is $(1 + \kappa)\ell$ -Lipschitz and

$$\|\nabla^2 P(x) - \nabla^2 P(x')\| \leq M_P \|x - x'\|, \quad M_P := 4\sqrt{2}\kappa^3\rho. \quad (2.21)$$

These properties follow from Luo et al. [[1](#), Lemmas 1–3].

The residual gives the distance estimate

$$\|y - y^*(x)\| \leq \frac{\|\nabla_y f(x, y)\|}{\mu}. \quad (2.22)$$

This follows from strong concavity and Cauchy–Schwarz. This estimate is used in the analysis, not as an executable certificate when μ is unknown.

For the inner objective $q_x(y) := -f(x, y)$, [\(2.16\)](#) gives $\mu \leq a(q_x; u, z) \leq \ell$ whenever $u \neq z$. This observable secant initializes persistent SCAR and scales the validation tolerance without evaluating either unknown parameter.

3 A parameter-free minimax algorithm

We instantiate [Algorithm 1](#) and [Theorem 2.1](#) with the value function $F = P$, using persistent SCAR to realize the working and validation oracles. There are two distinct tasks: maintaining a sufficiently accurate working oracle along outer steps, and validating a possible stationary point. The first uses a step-dependent number of residual halvings; the second requests a smaller residual only when the outer step is interior and the corrected gradient is already small.

3.1 Corrected value-function oracles

At an approximate maximizer y , define the Newton-corrected value

$$\hat{P}(x, y) := f(x, y) - \frac{1}{2} \nabla_y f(x, y)^T (\nabla_{yy}^2 f(x, y))^{-1} \nabla_y f(x, y). \quad (3.1)$$

Since $\nabla_{yy}^2 f \prec 0$, the correction raises the raw payoff and is exact for a concave quadratic inner problem. The corrected gradient and Hessian estimators are

$$\hat{g}(x, y) := \nabla_x f(x, y) - \nabla_{xy}^2 f(x, y) (\nabla_{yy}^2 f(x, y))^{-1} \nabla_y f(x, y), \quad (3.2a)$$

$$\hat{H}(x, y) := \mathcal{H}(x, y). \quad (3.2b)$$

Unlike the raw plug-in gradient used in Luo et al. [[1](#), Algorithm 2], [\(3.2a\)](#) cancels the linear error in the approximate maximizer. The Hessian estimator remains the Schur complement.

Lemma 3.1 (Corrected value and derivative errors). *Under [Assumption 2.2](#), for every x and y ,*

$$\|\hat{g}(x, y) - \nabla P(x)\| \leq \frac{(1 + \kappa)\rho}{2} \|y - y^*(x)\|^2, \quad (3.3a)$$

$$\|\hat{H}(x, y) - \nabla^2 P(x)\| \leq (1 + \kappa)^2 \rho \|y - y^*(x)\|. \quad (3.3b)$$

If $\|y - y^*(x)\| \leq \mu/\rho$, then

$$|\hat{P}(x, y) - P(x)| \leq \frac{7\rho}{24} \|y - y^*(x)\|^3. \quad (3.4)$$

The proof is given in [Section 4.1](#). The corrected value need not lie on a particular side of $P(x)$. Consequently, the acceptance analysis must account for the error of the stored proxy, rather than treat it as a certified function value.

3.2 Working accuracy and residual tracking

For $0 < \epsilon < 1$, set

$$\mathcal{A}_\epsilon := \log(e/\epsilon), \quad \sigma_0 := \mathcal{A}_\epsilon^{-1}, \quad c_0 := 2^{-12}, \quad r_\epsilon(\sigma) := \frac{\sqrt{\epsilon}}{4\sigma}. \quad (3.5)$$

The quantity $r_\epsilon(\sigma)$ sets the inner tracking accuracy and equals the radius at the gradient floor; the outer radius may be larger. A tuple $(\hat{P}, \hat{g}, \hat{H})$ computed at (x, y) is called working-accurate at scale σ when

$$\|\nabla_y f(x, y)\| \leq c_0 \ell r_\epsilon(\sigma). \quad (3.6)$$

The unknown constant ℓ appears only in this analytical invariant, not in any test performed by the algorithm.

If the invariant holds before a step of length r , then

$$N_{\text{tr}}(r, \sigma) := \left\lceil \log_2 \left(1 + \frac{r}{c_0 r_\epsilon(\sigma)} \right) \right\rceil \quad (3.7)$$

successful persistent SCAR halvings at the candidate outer point restore it. One additional halving at an unchanged outer point restores it when σ is doubled. These statements follow from joint gradient Lipschitz continuity; the proof is in [Lemma 4.1](#). In particular, the number of required halvings is computed without ℓ or μ . Initialization uses the observable secant $a_0 \leq \ell$ to establish the invariant.

3.3 SCAR realization of the oracle interface

The working tuple is $(\hat{P}, \hat{g}, \hat{H})$ from (3.1)–(3.2), together with its inner point y and cached residual gradient. The general UTR model (2.8) uses this tuple without modification. The next algorithm specifies the three working-query operations and the validator; the outer steps, acceptance tests, and scale updates are exactly those of [Algorithm 1](#). To write the oracle operations compactly, let

$$\begin{aligned} \text{Half}_x^0(y) &:= y, \\ \text{Half}_x^{j+1}(y) &:= \text{SCARHalf}(q_x, \text{Half}_x^j(y); \nu, \mathcal{M}), \\ \text{Target}_x(y, \tau) &:= \text{SCARTarget}(q_x, y, \tau; \nu, \mathcal{M}), \\ \mathcal{C}(x, y) &:= (\hat{P}(x, y), \hat{g}(x, y), \hat{H}(x, y); y, \nabla_y f(x, y)). \end{aligned} \quad (3.8)$$

The halving recursion means sequential calls, each using the calibration left by its predecessor; all calls, including target calls, share the same $(\nu, \mathcal{M})$. In $\mathcal{C}$, the auxiliary data are the inner point and its cached residual gradient; $y(w)$ denotes the inner point in w . Let e_1 be the first coordinate vector in $\mathbb{R}^{d_y}$, so every secant below uses two distinct points. Finally, $\text{UTR}(x_0, \epsilon, \sigma_0; \mathcal{W}, \mathcal{V})$ denotes [Algorithm 1](#).

Algorithm 2: Parameter-free minimax UTR with persistent SCAR

Input: $x_0 \in \mathbb{R}^{d_x}$, $y_{-1} \in \mathbb{R}^{d_y}$, $0 < \epsilon < 1$

- 1 $(\sigma_0, c_0) \leftarrow (1/\log(e/\epsilon), 2^{-12})$
- 2 $a_0 \leftarrow a(q_{x_0}; y_{-1}, y_{-1} + e_1)$; $(\nu, \mathcal{M}) \leftarrow (a_0, a_0)$
- 3 $\mathcal{W}_{\sigma_0}^0(x_0) := \mathcal{C}(x_0, \text{Target}_{x_0}(y_{-1}, c_0 a_0 r_\epsilon(\sigma_0)))$
- 4 $\mathcal{W}_{\sigma, z}(x; w) := \mathcal{C}(x, \text{Half}_x^{N_{\text{tr}}(\|x-z\|, \sigma)}(y(w)))$
- 5 $\mathcal{W}_\sigma^r(x; w) := \mathcal{C}(x, \text{Half}_x^1(y(w)))$
- 6 **function** $\mathcal{V}_\sigma(x; w)$:
 - 7 $y \leftarrow y(w)$; $a_v \leftarrow a(q_x; y, y + e_1)$
 - 8 $y_v \leftarrow \text{Target}_x(y, c_0 a_v \epsilon^{3/2})$
 - 9 **return** $(\hat{g}(x, y_v), \hat{H}(x, y_v))$
- 10 **end**
- 11 **return** $\text{UTR}(x_0, \epsilon, \sigma_0; \mathcal{W}, \mathcal{V})$

Lines 3–5 define routines: their right-hand sides are evaluated only when the outer method calls the corresponding oracle. The calibration initialized at line 2 is shared by every such evaluation, including rejected candidates and failed validations. The secant at line 7 is computed once and held fixed throughout line 8. The trial and refresh definitions distinguish zero-length trials ($N_{\text{tr}}(0, \sigma) = 0$) from scale refreshes (one halving). The outer assignments in lines 13 and 18 of Algorithm 1 ensure that a refresh starts from the old stored $y(w)$, not the rejected candidate or validation point. All residual and secant queries, with residual gradients cached when available, are included in the inner-gradient count.

3.4 Complexity guarantee

Theorem 3.1 (Class-uniform gradient-scaled complexity). *Fix $\ell \geq \mu > 0$ and $\rho > 0$. There exists $\epsilon_0 = \epsilon_0(\ell, \mu, \rho) \in (0, 1)$ such that the following holds for every twice continuously differentiable payoff $f : \mathbb{R}^{d_x} \times \mathbb{R}^{d_y} \rightarrow \mathbb{R}$ with ℓ -Lipschitz joint gradient, ρ -Lipschitz joint Hessian, μ -strongly concave inner sections, and a bounded-below value function $P(x) = \max_y f(x, y)$, and for every initialization (x_0, y_{-1}) . Set $\kappa = \ell/\mu$, $M_P = 4\sqrt{2}\kappa^3\rho$, and $\Delta_P = P(x_0) - \inf_x P(x)$. For every $0 < \epsilon \leq \epsilon_0$, run Algorithm 2 with exact evaluations of the derivatives of f used in its corrected oracles and globally solved trust-region subproblems. It terminates without using $\ell, \mu, \rho, \kappa, M_P$, or Δ_P as an input. Its output satisfies*

$$\|\nabla P(x_{\text{out}})\| < \epsilon, \quad \lambda_{\min}(\nabla^2 P(x_{\text{out}})) \geq -\frac{85}{16\sqrt{6}}\sqrt{M_P\epsilon}. \quad (3.9)$$

If Q counts all solved trust-region subproblems and N_y counts all inner-gradient evaluations, including the secant evaluations, then

$$Q = O\left(\Delta_P \kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2}\right) + O_{\text{fixed}}(\log^3(e/\epsilon)), \quad (3.10a)$$

$$N_y = O\left(\Delta_P \kappa^2 \sqrt{\rho} \epsilon^{-3/2}\right) + O_{\text{fixed}}(\sqrt{\kappa} \log^3(e/\epsilon)). \quad (3.10b)$$

Here O_{fixed} may depend on the fixed payoff, regularity constants, and initialization, but not on ϵ . Using internal tolerance $(1536/7225)\epsilon$ returns an ϵ -SOSP of P , with $M = M_P$ in Definition 2.2, whenever that internal tolerance is at most ϵ_0 .

Remark 3.1 (Meaning of parameter freedom). *The same $\epsilon_0(\ell, \mu, \rho)$ works for the entire parameter class and every initialization. The constants in the additive O_{fixed} terms need not be uniform over that class. The algorithm neither knows this threshold nor certifies that $\epsilon \leq \epsilon_0$. Thus the theorem is a class-uniform asymptotic guarantee, not an observable certificate for every input accuracy. Sufficient inequalities defining such a threshold are given in (4.6). The residual-only obstruction and its relation to the secant-scaled validator are discussed in Section 5.*

Role of the general UTR theorem. The corrected working and validation tuples satisfy [Assumption 2.1](#) below a class-uniform accuracy threshold. Thus the outer guarantee follows directly from [Theorem 2.1](#); no separate minimax descent or scale-search argument is needed. The accompanying estimates in [Proposition A.1](#) give a cubic-path bound that pays for tracking on accepted steps, while the stored-value bound controls tracking at failed trials. Persistent SCAR shares one curvature calibration across the entire run. The remaining work is the initialization, scale-refresh, and candidate-validation cost. [Section 4](#) gives all details.

4 NC–SC convergence analysis

The outer bounds are supplied by [Theorem 2.1](#) and [proposition A.1](#). We now verify the SCAR oracle errors and bound the inner work.

4.1 Corrected value-function oracles

Proof of [Lemma 3.1](#). Corrected value. Fix x and write $C = \nabla_{yy}^2 f(x, y)$, $q = \nabla_y f(x, y)$, and $\delta = y - y^*(x)$. Taylor expansion about y gives

$$v := q - C\delta, \quad \|v\| \leq \frac{\rho}{2}\|\delta\|^2, \quad P(x) = f(x, y) - q^T \delta + \frac{1}{2}\delta^T C \delta + R, \quad |R| \leq \frac{\rho}{6}\|\delta\|^3.$$

Using $q = C\delta + v$ and the definition of $\hat{P}$ yields

$$P(x) - \hat{P}(x, y) = R + \frac{1}{2}v^T C^{-1}v.$$

Strong concavity implies $\|C^{-1}\| \leq 1/\mu$. If $\|\delta\| \leq \mu/\rho$, then

$$\frac{1}{2}|v^T C^{-1}v| \leq \frac{\rho^2}{8\mu}\|\delta\|^4 \leq \frac{\rho}{8}\|\delta\|^3.$$

Adding the remainder proves [\(3.4\)](#).

Corrected gradient. Let

$$y_t := y^*(x) + t(y - y^*(x)), \quad t \in [0, 1].$$

Since f is twice continuously differentiable, the fundamental theorem of calculus applied to the two partial gradients along this segment gives the exact identities

$$\begin{aligned} \nabla_x f(x, y) - \nabla_x f(x, y^*(x)) &= \int_0^1 \nabla_{xy}^2 f(x, y_t)(y - y^*(x)) dt, \\ \nabla_y f(x, y) &= \int_0^1 \nabla_{yy}^2 f(x, y_t)(y - y^*(x)) dt, \end{aligned}$$

where the second identity uses $\nabla_y f(x, y^*(x)) = 0$. Substituting these identities into [\(3.2a\)](#) and using [\(2.20a\)](#), we obtain

$$\begin{aligned} \hat{g}(x, y) - \nabla P(x) &= \int_0^1 \left[\nabla_{xy}^2 f(x, y_t) - \nabla_{xy}^2 f(x, y) \right. \\ &\quad \left. + \nabla_{xy}^2 f(x, y) (\nabla_{yy}^2 f(x, y))^{-1} (\nabla_{yy}^2 f(x, y) - \nabla_{yy}^2 f(x, y_t)) \right] \\ &\quad \times (y - y^*(x)) dt. \end{aligned}$$

This identity displays the cancellation of the linear inner error. Hessian Lipschitzness, together with

$$\left\| \nabla_{xy}^2 f(x, y) (\nabla_{yy}^2 f(x, y))^{-1} \right\| \leq \kappa,$$

bounds the norm of the integrand by

$$(1 + \kappa)\rho(1 - t)\|y - y^*(x)\|^2.$$

Integrating over $t \in [0, 1]$ proves (3.3a).

Corrected Hessian. Retain $C = \nabla_{yy}^2 f(x, y)$ and set

$$B := \nabla_{xy}^2 f(x, y), \quad B_* := \nabla_{xy}^2 f(x, y^*(x)), \quad C_* := \nabla_{yy}^2 f(x, y^*(x)).$$

The inverse identity gives

$$\begin{aligned} & \left\| (\nabla_{yy}^2 f(x, y))^{-1} - (\nabla_{yy}^2 f(x, y^*(x)))^{-1} \right\| \\ & \leq \frac{\rho}{\mu^2} \|y - y^*(x)\|. \end{aligned}$$

The Schur-complement products satisfy the exact telescoping identity

$$\begin{aligned} BC^{-1}B^T - B_*C_*^{-1}B_*^T &= (B - B_*)C^{-1}B^T + B_*(C^{-1} - C_*^{-1})B^T \\ &\quad + B_*C_*^{-1}(B - B_*)^T. \end{aligned}$$

Hessian Lipschitzness, $\|B\|, \|B_*\| \leq \ell$, and $\|C^{-1}\|, \|C_*^{-1}\| \leq 1/\mu$ bound these three terms by $\kappa\rho\|y - y^*(x)\|$, $\kappa^2\rho\|y - y^*(x)\|$, and $\kappa\rho\|y - y^*(x)\|$, respectively. Adding the xx -block error gives

$$\begin{aligned} \|\hat{H}(x, y) - \nabla^2 P(x)\| &\leq \rho\|y - y^*(x)\| + 2\frac{\ell}{\mu}\rho\|y - y^*(x)\| \\ &\quad + \frac{\ell^2}{\mu^2}\rho\|y - y^*(x)\| \\ &= (1 + \kappa)^2\rho\|y - y^*(x)\|, \end{aligned}$$

which proves (3.3b). □

4.2 Tracking and oracle accuracy

Lemma 4.1 (Variable-step tracking). *Under Assumption 2.2, suppose (3.6) holds at (x, y) . For any step d of length r , applying $N_{\text{tr}}(r, \sigma)$ successful persistent SCAR halvings to q_{x+d} , starting from y , produces a working-accurate tuple at $x + d$ with the same scale. At an unchanged x , one successful halving restores the invariant after σ is doubled.*

Proof of Lemma 4.1. Lipschitz continuity of the full gradient gives

$$\|\nabla_y f(x + d, y)\| \leq \|\nabla_y f(x, y)\| + \ell r \leq \ell(c_0 r_\epsilon(\sigma) + r).$$

After $N_{\text{tr}}(r, \sigma)$ successful halvings, this is at most $c_0 \ell r_\epsilon(\sigma)$ by (3.7). The case $r = 0$ requires no halving. At an unchanged x , use $r_\epsilon(2\sigma) = r_\epsilon(\sigma)/2$. □

The error constants used only in the analysis are

$$C_g := \frac{c_0^2(1 + \kappa)\rho\kappa^2}{32}, \quad C_v := \frac{7c_0^3\rho\kappa^3}{1536}. \quad (4.1)$$

We now verify the assumptions of the general UTR theorem.

Lemma 4.2 (Working and validation oracle accuracy). *Fix common constants (ℓ, μ, ρ) in [Assumption 2.2](#). For all sufficiently small $\epsilon > 0$, with a threshold depending only on these constants, every working-accurate tuple at $\sigma \geq \sigma_0 = \mathcal{A}_\epsilon^{-1}$ satisfies*

$$\|\hat{g} - \nabla P(x)\| \leq C_g \frac{\epsilon}{\sigma^2}, \quad |\hat{P} - P(x)| \leq C_v \frac{\epsilon^{3/2}}{\sigma^3}. \quad (4.2)$$

If also $\sigma^2 \geq M_P/6$, then

$$\begin{aligned} \|\hat{g} - \nabla P(x)\| &\leq \epsilon/1024, \\ \|\hat{H} - \nabla^2 P(x)\| &\leq \sigma\sqrt{\epsilon}/64, \\ |\hat{P} - P(x)| &\leq \epsilon^{3/2}/(4096\sigma). \end{aligned} \quad (4.3)$$

Furthermore, for any secant $a_x = a(q_x; u, z)$ with $u \neq z$, any point y_v satisfying

$$\|\nabla_y f(x, y_v)\| \leq c_0 a_x \epsilon^{3/2} \quad (4.4)$$

obeys, at every $\sigma \geq \sigma_0$,

$$\begin{aligned} \|\hat{g}(x, y_v) - \nabla P(x)\| &\leq \epsilon/1024, \\ \|\hat{H}(x, y_v) - \nabla^2 P(x)\| &\leq \sigma\sqrt{\epsilon}/32. \end{aligned} \quad (4.5)$$

Consequently [Assumption 2.1](#) holds with $F = P$, $M = M_P$, and the displayed C_g, C_v . All conclusions hold uniformly over the parameter class, the outer and inner points, and the choices of secants.

Proof of [Lemma 4.2](#). It is sufficient to restrict $\epsilon \in (0, 1)$ so that

$$\begin{aligned} \sqrt{\epsilon} \mathcal{A}_\epsilon &\leq \frac{4\mu}{c_0 \kappa \rho}, & \epsilon^2 &\leq \frac{1}{512(1+\kappa)\rho c_0^2 \kappa^2}, \\ \epsilon \mathcal{A}_\epsilon &\leq \frac{1}{32(1+\kappa)^2 \rho c_0 \kappa}, & \mathcal{A}_\epsilon &> \sqrt{\frac{6}{M_P}}. \end{aligned} \quad (4.6)$$

The first three left-hand sides tend to zero, and the last tends to infinity, as $\epsilon \downarrow 0$. Hence a positive threshold, depending only on (ℓ, μ, ρ) , makes these inequalities valid for all smaller positive ϵ . The last condition ensures $\sigma_0 < \sqrt{M_P/6}$, which will also be used in the termination proof.

Strong concavity and the working residual invariant give

$$e := \|y - y^*(x)\| \leq \frac{c_0 \kappa \sqrt{\epsilon}}{4\sigma} \leq \frac{c_0 \kappa}{4} \sqrt{\epsilon} \mathcal{A}_\epsilon \leq \frac{\mu}{\rho}.$$

The corrected-oracle bounds in [Lemma 3.1](#) yield

$$\|\hat{g} - \nabla P(x)\| \leq \frac{(1+\kappa)\rho}{2} e^2, \quad |\hat{P} - P(x)| \leq \frac{7\rho}{24} e^3.$$

Substituting the first bound on e proves [\(4.2\)](#).

At a safe scale $\sigma^2 \geq M_P/6$, use $M_P = 4\sqrt{2}\kappa^3\rho$ and $\kappa \geq 1$ to obtain

$$\begin{aligned} \|\hat{g} - \nabla P(x)\| &\leq \frac{3c_0^2(1+\kappa)}{64\sqrt{2}\kappa} \epsilon \leq \frac{3c_0^2}{32\sqrt{2}} \epsilon, \\ \|\hat{H} - \nabla^2 P(x)\| &\leq \frac{3c_0(1+\kappa)^2}{8\sqrt{2}\kappa^2} \sigma\sqrt{\epsilon} \leq \frac{3c_0}{2\sqrt{2}} \sigma\sqrt{\epsilon}, \\ |\hat{P} - P(x)| &\leq \frac{7c_0^3}{1024\sqrt{2}} \frac{\epsilon^{3/2}}{\sigma}. \end{aligned}$$

Since $c_0 = 2^{-12}$, these imply [\(4.3\)](#).

For validation, the secant bound $a_x \leq \ell$ and (4.4) give

$$\|y_v - y^*(x)\| \leq c_0 \kappa \epsilon^{3/2}.$$

The global derivative bounds in Lemma 3.1 then imply

$$\|\widehat{g}(x, y_v) - \nabla P(x)\| \leq \frac{(1 + \kappa) \rho c_0^2 \kappa^2}{2} \epsilon^3 \leq \frac{\epsilon}{1024},$$

by the second condition in (4.6). Similarly, its third condition and $\sigma \geq \mathcal{A}_\epsilon^{-1}$ give

$$\|\widehat{H}(x, y_v) - \nabla^2 P(x)\| \leq (1 + \kappa)^2 \rho c_0 \kappa \epsilon^{3/2} \leq \frac{\sqrt{\epsilon}}{32 \mathcal{A}_\epsilon} \leq \frac{\sigma \sqrt{\epsilon}}{32}.$$

This proves (4.5). None of the restrictions uses the payoff, initialization, evaluation points, or chosen secants. $\square$

4.3 Direct application and total inner work

Proof of Theorem 3.1. Choose $\epsilon_0 \in (0, 1)$ so that (4.6) holds for every $0 < \epsilon \leq \epsilon_0$. By Lemma 4.2, the corrected working tuples and the SCAR validator satisfy Assumption 2.1 with $F = P$, $M = M_P$, and the constants C_g, C_v in (4.1). Initialization and Lemma 4.1 maintain the working residual invariant. Every inner sequence terminates by Lemma 2.2 and positivity of the requested residual targets. Thus Theorem 2.1 applies directly to Algorithm 2. The onset restrictions depend only on the common constants (ℓ, μ, ρ) , and they also ensure $\sigma_0 = \mathcal{A}_\epsilon^{-1} < \sqrt{M_P/6}$. Consequently the scale bound in Proposition A.1 gives $\bar{\sigma} = 2\sqrt{M_P/6}$. Throughout this proof, K, J, h_0 and B_ϵ are the counts and auxiliary quantities defined in that proposition and (A.1), specialized to $F = P$.

We first specialize the budget B_ϵ in (A.1). Since $\sigma_0 = \mathcal{A}_\epsilon^{-1}$, its value-error term is $(23/7)C_v \epsilon^{3/2} \mathcal{A}_\epsilon^3$. At the initial stored tuple, (4.2) gives

$$\|\widehat{g}_0\| \leq \|\nabla P(x_0)\| + C_g \epsilon \mathcal{A}_\epsilon^2.$$

For each fixed payoff and initialization, this implies

$$\lceil \log_2(h_0/\epsilon) \rceil = O_{\text{fixed}}(\mathcal{A}_\epsilon), \quad 1 + \log_2 \left(1 + \frac{5C_g}{4\sigma_0^2} \right) = O_{\text{fixed}}(1 + \log \mathcal{A}_\epsilon),$$

where $h_0 = \max\{\|\widehat{g}_0\|, \epsilon\}$. Because $\epsilon^{3/2}/\sigma_0 = \epsilon^{3/2} \mathcal{A}_\epsilon$, substitution into (A.1) yields

$$B_\epsilon = \Delta_P + O_{\text{fixed}}(\epsilon^{3/2} \mathcal{A}_\epsilon^3). \quad (4.7)$$

The count bound (A.2), together with $\bar{\sigma} = 2\sqrt{M_P/6}$, now proves (3.10a). The output guarantee (2.11) gives (3.9) with the same scale bound. The tolerance conversion follows from $(85/(16\sqrt{6}))^2 = 7225/1536$.

It remains to count successful halvings. On an accepted step, set $z_k = \sigma_k r_k / \sqrt{\epsilon}$. Since $c_0 = 2^{-12}$,

$$N_{\text{tr},k} = \lceil \log_2(1 + 2^{14} z_k) \rceil \leq 16 + z_k^3.$$

For $0 \leq z \leq 1$, the ceiling is at most 15. For $z \geq 1$, use $1 + 2^{14} z \leq 2^{15} z$ and $\log_2 z \leq z^3$. Thus (A.2) and (A.3) give

$$\begin{aligned} \sum_{\text{accepted}} N_{\text{tr},k} &\leq 16K + \epsilon^{-3/2} \sum_{\text{accepted}} \sigma_k^3 r_k^3 \\ &\leq 16K + 8\bar{\sigma} B_\epsilon \epsilon^{-3/2} \\ &\leq 16392\bar{\sigma} B_\epsilon \epsilon^{-3/2}. \end{aligned} \quad (4.8)$$

To bound tracking at failed trials, put $L_P = (1 + \kappa)\ell$. Smoothness and the lower bound P^* imply

$$\|\nabla P(x)\|^2 \leq 2L_P(P(x) - P^*).$$

At every stored iterate, (A.3) and (4.2) therefore give

$$h \leq \bar{h}_\epsilon := \max \left\{ \epsilon, \sqrt{2L_P B_\epsilon} + C_g \epsilon \mathcal{A}_\epsilon^2 \right\} = O_{\text{fixed}}(1).$$

Since $r \leq R(h, \sigma)$, each failed trial requires at most

$$N_{\text{fail}} := \left\lceil \log_2 \left(1 + \frac{1}{c_0} \sqrt{\frac{\bar{h}_\epsilon}{\epsilon}} \right) \right\rceil = O_{\text{fixed}}(\mathcal{A}_\epsilon)$$

working halvings. A terminal trial has $h = \epsilon$ and $r < r_\epsilon(\sigma)$, so its working tracking uses at most $\lceil \log_2(1 + 1/c_0) \rceil = 13$ halvings. Each failure uses one additional halving for the refresh.

Immediately before a validation, the residual is at most $c_0 \ell \sqrt{\epsilon}/(4\sigma)$. The validation target is at least $c_0 \mu \epsilon^{3/2}$, so the number of additional halvings is at most

$$N_{\text{val}} := \left\lceil \log_2 \max \left\{ 1, \frac{\kappa \mathcal{A}_\epsilon}{4\epsilon} \right\} \right\rceil = O_{\text{fixed}}(\mathcal{A}_\epsilon).$$

There are at most $J + 1$ validations. Initialization uses at most

$$N_{\text{init}} := \left\lceil \log_2 \max \left\{ 1, \frac{\|\nabla_y f(x_0, y_{-1})\|}{c_0 a_0 r_\epsilon(\sigma_0)} \right\} \right\rceil = O_{\text{fixed}}(\mathcal{A}_\epsilon)$$

halvings. Consequently the total number of successful halvings obeys

$$N_{\text{half}} \leq \sum_{\text{accepted}} N_{\text{tr},k} + J N_{\text{fail}} + J + 13 + (J + 1) N_{\text{val}} + N_{\text{init}}. \quad (4.9)$$

By (A.2), $J = O_{\text{fixed}}(1 + \log \mathcal{A}_\epsilon)$. Equations (4.7) and (4.8) therefore imply

$$N_{\text{half}} = O(\Delta_P \sqrt{M_P} \epsilon^{-3/2}) + O_{\text{fixed}}(\mathcal{A}_\epsilon^3).$$

All inner objectives $q_x = -f(x, \cdot)$ share the same (μ, ℓ) . Apply Lemma 2.2 once to the complete sequence, including rejected candidates, refreshes, and validators. The initial and validation secants require at most $2(J + 2)$ additional gradient evaluations, giving

$$N_y = O(\sqrt{\kappa} [N_{\text{half}} + \lceil \log_4 \kappa \rceil] + J + 1). \quad (4.10)$$

This proves (3.10b). Finally, $\sqrt{M_P} = \Theta(\kappa^{3/2} \sqrt{\rho})$ and $\sqrt{\kappa M_P} = \Theta(\kappa^2 \sqrt{\rho})$ give the stated leading dependence. $\square$

5 Scope of the parameter-free guarantee

This section elaborates on Remark 3.1. The distinction is between an algorithm that needs no problem parameters and a finite-accuracy certificate that can itself be verified without those parameters.

5.1 The residual-only obstruction

A parameter-free inner method such as SCAR can return a point satisfying the prescribed and directly verifiable condition $\|\nabla_y f(x, y)\| \leq \tau$. This fact alone does not provide a certificate for the value or gradient of $P(x) = \max_y f(x, y)$ that is uniform over problem classes having no positive common lower bound on the strong-concavity modulus. The issue is not merely a loss in the complexity bound: an outer stopping test based only on a residual-corrected oracle can be false.

The following scalar family shows that this difficulty persists even for the corrected value (3.1). Let

$$c := \frac{1}{2}, \quad \psi(t) := \frac{t^2}{2} + c(1 - \cos t), \quad A := \frac{1 - c}{2\pi^2},$$

and, for $0 < \mu \leq 1/4$, define

$$f_\mu(x, y) := A \tanh(x/A) (1 - \cos(2\pi\sqrt{\mu}y)) - \psi(2\sqrt{\mu}y - 1). \quad (5.1)$$

Because $1 - c \leq \psi''(t) \leq 1 + c$,

$$\partial_{yy}^2 f_\mu(x, y) = 4A\pi^2\mu \tanh(x/A) \cos(2\pi\sqrt{\mu}y) - 4\mu\psi''(2\sqrt{\mu}y - 1) \leq -2(1 - c)\mu = -\mu.$$

Thus $f_\mu(x, \cdot)$ is globally μ -strongly concave. Its second derivatives are uniformly bounded for $0 < \mu \leq 1/4$. Its third derivatives are also uniformly bounded: the four distinct block types have bounds $12/A^2$, $2\pi/A$, π^2 , and $A\pi^3 + c$, respectively. Hence the joint gradient-smoothness and Hessian-Lipschitz constants admit finite upper bounds independent of μ . Moreover,

$$P_\mu(x) \geq f_\mu\left(x, \frac{1}{2\sqrt{\mu}}\right) = 2A \tanh(x/A) \geq -2A,$$

so the value function is bounded below. At $x = 0$ and $y = 1/(2\sqrt{\mu})$, $-\partial_{yy}^2 f_\mu(x, y) = 4\mu\psi''(0) = 6\mu$. Consequently, the largest valid global strong-concavity modulus lies between μ and 6μ and genuinely tends to zero along the family.

At $x = 0$, the exact maximizer is $y_\mu^*(0) = 1/(2\sqrt{\mu})$. At the point $y = 0$, however,

$$|\partial_y f_\mu(0, 0)| = 2\sqrt{\mu}|\psi'(-1)| = (2 + \sin 1)\sqrt{\mu} \longrightarrow 0, \quad \hat{g}_\mu(0, 0) = 0, \quad \hat{H}_\mu(0, 0) = 0.$$

In contrast, Danskin's theorem gives

$$P'_\mu(0) = \partial_x f_\mu\left(0, \frac{1}{2\sqrt{\mu}}\right) = 1 - \cos \pi = 2.$$

The corrected value is wrong by a fixed amount as well. Namely, $P_\mu(0) = 0$, whereas

$$\hat{P}_\mu(0, 0) = -\psi(-1) + \frac{\psi'(-1)^2}{2\psi''(-1)} \approx 0.0647, \quad (5.2)$$

independently of μ . Therefore a residual tending to zero does not uniformly control either the corrected value or the corrected gradient over the union of these parameter classes. The smoothness constants remain bounded while the best strong-concavity modulus tends to zero.

Consequently, choosing SCAR as a residual module removes the inner solver's dependence on ℓ and μ , but the residual alone does not yield a single certificate valid across all positive strong-concavity moduli. Nor does a prescribed finite sequence of positive tolerances repair the issue. For any finite list $\tau_0, \dots, \tau_J > 0$, choose

$$0 < \mu < \min \left\{ \frac{1}{4}, \left(\frac{\min_j \tau_j}{2 + \sin 1} \right)^2 \right\}$$

and repeat each inner call at the outer point $x = 0$, initialized at $y = 0$. Every residual test is then already satisfied and may return the same false outer derivatives. This observation covers any predetermined finite number of tolerance-halving phases. It does not claim that indefinite refinement cannot eventually move the inner iterate; it shows that a finite prescribed refinement depth is not a certificate.

Once a positive common lower bound μ is fixed, the condition

$$\tau < (2 + \sin 1)\sqrt{\mu}$$

excludes $y = 0$ as a legal return of an inner call with residual target τ . More generally, every legal return satisfies $\|y - y^*(x)\| \leq \tau/\mu$. Thus the family illustrates why a residual-to-oracle estimate must retain its dependence on the common strong-concavity bound. It does not rule out eventual accuracy for any fixed problem.

This example concerns prescribed absolute residual targets. It does not establish failure of [Algorithm 2](#), whose working and validation targets use observable secants and whose accepted inner points are tracked persistently. [Theorem 3.1](#) asserts uniformity after (ℓ, μ, ρ) have been fixed; it makes no claim that one threshold works over their union as $\mu \downarrow 0$.

5.2 Validation and stored-oracle bookkeeping

SCAR returns a verified residual, not a certified lower bound on μ or an interval containing $P(x)$. Its line-search estimate satisfies the tested inequalities but is not a certified global smoothness bound. A successful residual halving likewise does not certify that its current strong-convexity guess lies below the true modulus. In the minimax proof, the common regularity constants are used to establish the working and validation error bounds, even though they are absent from the executable algorithm.

The corrected value $\hat{P}$ is neither a guaranteed upper nor a guaranteed lower bound on P . An accepted trial at an unsafe scale need not decrease P . The proof instead telescopes the stored potential used in the proof of [Theorem 2.1](#) and pays for every positive jump caused by a scale refresh; see [Lemma A.4](#). This requires three implementation rules: the scale is nondecreasing, an accepted corrected tuple is retained without extra refinement, and a rejected candidate is discarded before the current tuple is refreshed. Refining a stored inner iterate without accounting for the resulting proxy change would invalidate that argument.

The validator is used only after an interior trial at $h = \epsilon$. Its residual target $c_0 a_x \epsilon^{3/2}$ controls inner accuracy; it is not a trust-region scale or a Hessian threshold. Under the class-uniform small-accuracy conditions, its tests imply true second-order stationarity. They do not reveal whether those conditions hold. Refining or doubling the requested accuracy alone therefore does not provide the missing observable certificate. Such a certificate would require an additional argument or information, for example a certified inner gap, a value-function interval, a known strong-concavity lower bound, or problem-specific structure.

6 Numerical Experiments

We study outer progress on a W-shaped problem (A), nonlinear correction and persistent inner tracking on a controlled path (B), and robust RBF classification on real data (C). The baselines are MCN [1], GRTR [4], and, in A, HSDA [5]. Full trajectories, search results, and diagnostics are retained in the accompanying experimental archive; [Section D](#) gives the essential reproduction protocol.

Implementation and evaluation. We use *parameter-free minimax UTR with persistent SCAR* ([Algorithm 2](#)), abbreviated *Minimax UTR* in this section. Its standard implementation uses ordinary persistent SCAR. We denote its early-return variant by *Minimax UTR-early*: it shares the outer driver but returns from an AR attempt when an already computed gradient of the original inner objective has halved the entry residual. Both retain the curvature state and use corrected oracles. MCN and GRTR use the raw gradient f_x and Schur Hessian $f_{xx} - f_{xy}f_{yy}^{-1}f_{yx}$. Both UTR variants use fixed settings: $c_0 = 1/128$ and $\sigma_0 = 1/\log(e/\epsilon)$, without inputs (ℓ, μ, ρ) or instance-specific calibration. The experimental c_0 differs from 2^{-12} in [Algorithm 2](#); these implementations, including early return, are not claimed to inherit automatically every guarantee of [Theorem 3.1](#).

We count solved outer subproblems Q and full-vector inner-gradient queries N_y , including initialization and performed residual, secant, and backtracking checks. UTR’s Q includes rejected trials and terminal validation. Hessian construction, linear algebra, and independent primal evaluation are separate costs. First hit means the first saved point meeting the stated true-primal target; function values do not determine success. Evaluation uses analytic quantities in A and independent high-accuracy inner solves in B/C, never fed back to the online methods. Search costs are reported separately. A/C compare configurations tuned on the same instance, not held-out performance; no statistical or cross-machine timing advantage is claimed.

6.1 Benchmark A: outer behavior on a W-shaped problem

Following Luo et al. [1], let

$$f(x, y) = w(x_3) + x_1y_1 - y_1^2/40 + x_2y_2 - 5y_2^2/2, \quad P(x) = 10x_1^2 + 0.1x_2^2 + w(x_3),$$

where, for $r = |t|$,

$$w(t) = \begin{cases} -0.1r^2 + r^3/3, & r \leq 0.1, \\ -0.01r + 0.001/3, & 0.1 < r \leq 0.5, \\ 0.1(r - 0.6)^2 + (r - 0.6)^3/3 - 2/375, & r > 0.5. \end{cases}$$

Here $y^*(x) = (20x_1, x_2/5)$ and $P^* = -2/375$. For the four starts in [Table 2](#), with $y_0 = 0$, the common target is

$$\|\nabla P(x)\| \leq 10^{-8}, \quad \lambda_{\min}(\nabla^2 P(x)) \geq -\sqrt{2} \cdot 10^{-8}. \quad (6.1)$$

Each baseline is tuned separately at each start, minimizing $(Q^{\text{hit}}, N_y^{\text{hit}})$ with a fixed tie order and 100-update budget. MCN/HSDA use 800/8,000 configurations per start; GRTR uses a staged search. Minimax UTR retains native stopping with a 2,000-subproblem safety budget. These are not equal-search-budget comparisons ([Section D](#)).

Both UTR variants match the smallest reported outer count at the first two starts and improve it at the other two. Their outer trajectories coincide, while early return sharply reduces full-SCAR work ([Figure 1](#)). Tuned baselines nevertheless use fewer inner queries at three starts. The quadratic inner makes the corrected value and gradient exact in exact arithmetic, so A tests outer behavior, not nonlinear tracking. Also, the cubic tails preclude a finite global joint gradient-Lipschitz constant; this literature benchmark does not verify every assumption of the minimax theorem.

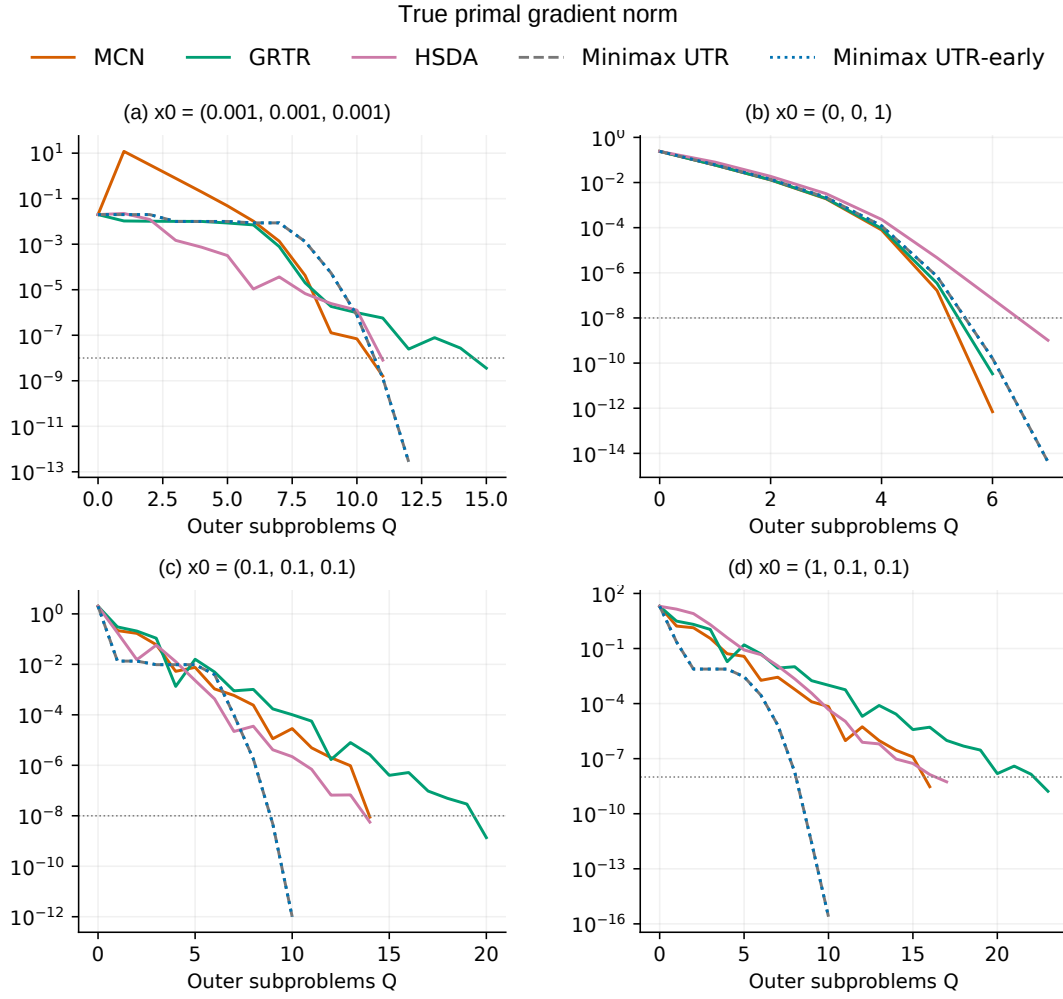


Figure 1: A: true gradient norm versus Q at all four starts. Baselines end at first hit; UTR retains native termination. Full/early UTR overlap. Actual nonmonotonicity is retained; no running minimum.

x_0	MCN	GRTR	HSDA	Minimax UTR	Minimax UTR-early
(.001, .001, .001)	11/440	15/300	11/440	11/23,784	11/1,528
(0, 0, 1)	6/24	6/30	7/28	6/8	6/8
(.1, .1, .1)	14/560	20/400	14/560	9/19,525	9/1,326
(1, .1, .1)	16/640	23/460	17/680	9/18,203	9/1,389

Table 2: A: $Q^{\text{hit}}/N_y^{\text{hit}}$, including initialization. Baseline parameters are selected separately for each row; UTR settings are fixed. These are first-hit, not native-termination, costs.

6.2 Benchmark B: correction and persistent tracking

For $x = (u, v)$, define the curved-valley payoff

$$\begin{aligned}
P_0(u, v) &= \sqrt{1 + (u - 35)^2} - 1 + 0.35[1 - \cos(0.7(u - 35))] + \log \cosh(v - h(u)), \\
h(u) &= 0.8 \sin(u/2) + 0.4 \sin(\sqrt{2}u/2), \quad C = 2^{-1/2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \\
f_D(x, y) &= P_0(x) + 0.1 \sum_{i=1}^2 [1 - \cos((Cx)_i + d_i y_i)] - 3 \|Dy\|^2,
\end{aligned} \tag{6.2}$$

where $D = \text{diag}(d_1, d_2)$ is invertible. The base instance uses $D = I$, $x_0 = y_0 = 0$, $\epsilon = 10^{-6}$, and bounds $(\ell, \mu, L_y, \rho, \bar{M}_P) = (6.2, 5.9, 6.1, 4.32283, 4.14516)$. The nonlinear inner maximizer satisfies $0.1 \sin((Cx)_i + z_i^*) = 6z_i^*$ for $z = Dy$; hence $|z_i^*| \leq 1/60$. Online derivatives require no root solve. Roots are used only for exact-primal controls and offline evaluation; $P \neq P_0$ and P^* is not assumed known.

Correction and outer-path reproduction. At $x = (.3, -.5)$, perturb $y^*(x)$ by $\delta(1, 1)/\sqrt{2}$ for seven scales from 10^{-4} to 10^{-1} . The raw-gradient, corrected-gradient, and Schur-Hessian error slopes are respectively 0.9992, 2.0055, and 1.0060, with a 70-digit cross-check (Figure 2). Thus gradient correction removes the linear error, whereas Hessian error remains first order.

Exact-primal UTR and Minimax UTR-early both reach $\|\nabla P\| \leq 10^{-6}$ and $\lambda_{\min} \geq -\sqrt{\bar{M}_P} 10^{-6}$ after 46 accepted updates and 48 subproblems. Early uses 779 inner queries; acceptance decisions agree and the maximum single-coordinate path difference is 1.16×10^{-9} . Exact MCN needs 153 updates with its prescribed $M = 28.3767$, versus 60/84 with $M = \bar{M}_P$ or $2\bar{M}_P$, respectively. These exact-oracle controls isolate outer scaling, not inner tracking; root computation is not free (Section D).

Fixed-path replay and restart ablation. Freeze exact UTR’s 47 accepted points and incoming scales σ_t . For $D_\gamma = \text{diag}(\gamma^{-1/2}, \gamma^{1/2})$, $\gamma \in \{1, 2, 4, 8\}$, the primal is unchanged. Recount all work from zero and require the same intrinsic target at every point:

$$\|D_\gamma^{-1} \nabla_y f_{D_\gamma}(x_t, y_t)\| \leq r_t, \quad r_t = \frac{c_0 \ell \sqrt{\epsilon}}{4\sigma_t}. \tag{6.3}$$

The external scheduler knows ℓ ; SCAR does not. Solvers update in y , without preconditioning, and AGD receives $L_{y,\gamma} = 6.1\gamma$, $\mu_\gamma = 5.9/\gamma$. Persistent SCAR-early retains $y, \nu, \mathcal{M}$; warm restart retains only y ; cold restart resets both. All checks and secants are charged.

At $\gamma = 1$, persistent full SCAR costs 17,021 queries versus 462 for early, a factor of 36.84. AGD is still cheaper, and persistence itself costs more than warm restart. At $\gamma = 8$, persistence instead saves a factor of 4.73 over warm restart. Downward curvature corrections $\nu \leftarrow \nu/4$ number 2/89/92 for persistent/warm/cold restart (1/44/46 at $\gamma = 4$). Only 46 of the 13,643 saved queries at $\gamma = 8$ are secant savings; the rest includes

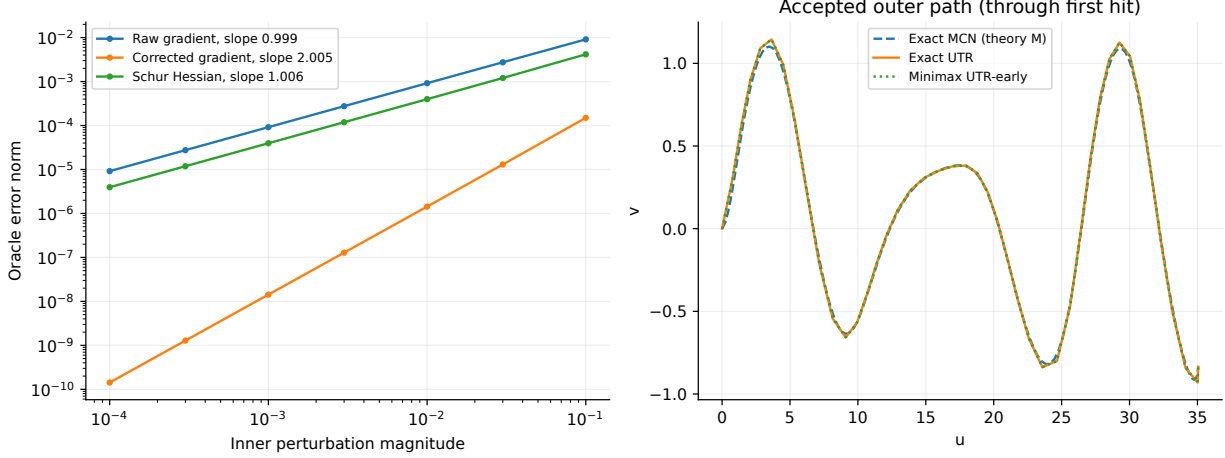


Figure 2: B: controlled inner-error scaling (left) and accepted outer paths (right). Exact UTR and Minimax UTR-early nearly coincide; exact MCN is an outer-scale control.

γ	$\bar{\kappa}_y$	Persistent	Warm restart	Cold restart	Warm AGD	Cold AGD
1	1.03	462	391	445	195	187
2	4.14	632	679	787	1,068	1,255
4	16.54	1,439	5,837	6,316	2,566	3,017
8	66.17	3,654	17,297	18,325	5,492	6,439

Table 3: B: fixed-path N_y^{path} at identical intrinsic targets. The three restart columns use SCAR-early; AGD knows the analytic conditioning bounds. The primal and outer path are unchanged.

all halving and checking work, not solely curvature identification. All 940 returned points satisfy the common target. This supports amortized state reuse on the controlled path, not uniform superiority to AGD or a fitted complexity exponent. Replay excludes rejected trials and outer validation, so 462 and the full-run cost 779 are distinct quantities.

6.3 Benchmark C: robust RBF classification

We use WDBC’s 569 samples and 30 features, with a seed-2026 stratified split into 455 training and 114 test samples. Standardization is fitted only on training data; test labels are not used for selection. With five trainable centers $u_j \in \mathbb{R}^{30}$, let

$$s_x(a) = \sum_{j=1}^5 \tanh(v_j) e^{-\|a - u_j\|^2/8} + \tanh(v_0), \quad x \in \mathbb{R}^{156}.$$

For labels $b_i \in \{-1, 1\}$ and unconstrained sample-wise perturbations $z_i = \sqrt{n} y_i$, $n = 455$, we minimize the maximum over $y \in \mathbb{R}^{455 \times 30}$ of

$$f(x, y) = \frac{1}{n} \sum_{i=1}^n \left[\log(1 + e^{-b_i s_x(a_i + \sqrt{n} y_i)}) - \|\sqrt{n} y_i\|^2 \right] + \frac{10^{-4}}{2} \|x\|^2. \quad (6.4)$$

The bandwidth and adversarial penalty are fixed at 2. Separability requires only 30×30 inner blocks and a 156×156 Schur Hessian. All methods use $y_0 = 0$ and the same frozen negative-curvature start, se-

Method	Configuration	Q^{hit}	N_y^{hit}	$\ \nabla P\ _{\text{hit}}$
Minimax UTR-early	Fixed	154	1,186	$3.19 \cdot 10^{-7}$
MCN	$M = 10^{-3}, \tau = 10^{-4}$	93	563	$9.06 \cdot 10^{-8}$
GRTR	$\tilde{L}_2 = 10^{-4}, \tau = 10^{-4}$	30	356	$9.26 \cdot 10^{-7}$

Table 4: C: initialization-inclusive first-hit costs. Baselines are selected on this same instance. All hit values satisfy $P \approx .493433306$, $\lambda_{\min} \approx 6.48 \cdot 10^{-4}$; this does not certify global optimality. Search costs are separate.

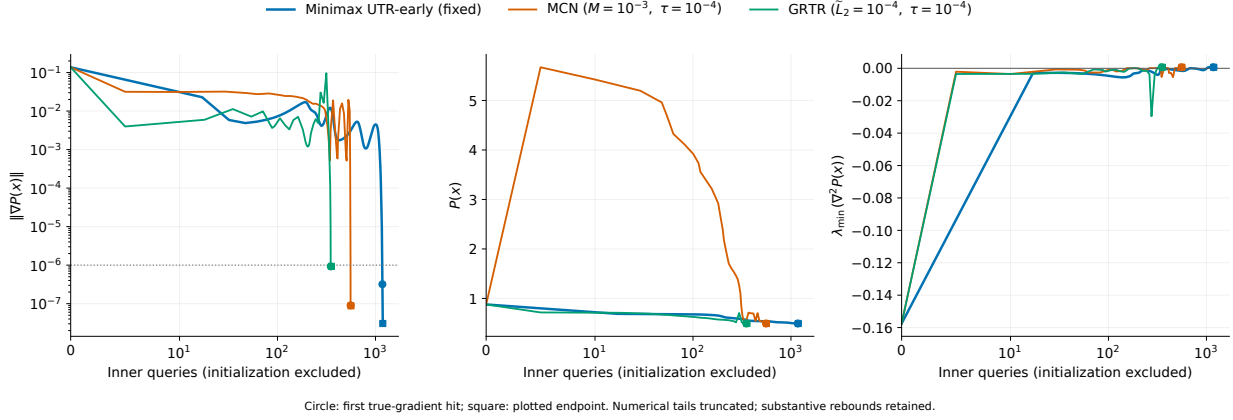


Figure 3: C: true gradient, objective, and minimum Hessian eigenvalue. Plots subtract initialization (10 queries for UTR, 6 per baseline); tables include it. Circles denote first hit, squares display endpoints. The symmetric-log axis retains zero. Only terminal numerical tails are trimmed; actual rebounds remain (Section D).

lected geometrically from 33 predetermined candidates, without algorithm outcomes: initially $P = .8783993$, $\|\nabla P\| = .1371588$, and $\lambda_{\min} = -.1574913$ (Section D).

Practical calibration. Both baselines use warm-start AGD with restarted momentum, $L_y = 3.25$, $\mu_y = .175188$, step $1/L_y$, and momentum $(\sqrt{L_y/\mu_y} - 1)/(\sqrt{L_y/\mu_y} + 1)$. We search

$$\begin{aligned}
M &\in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\} & (\text{MCN}), \\
\tilde{L}_2 &\in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\} & (\text{GRTR}), \\
\tau_{\text{inner}} &\in \{10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\} & (\text{both}).
\end{aligned} \tag{6.5}$$

Inner stopping requires $\|\nabla_y f\| \leq \tau_{\text{inner}}$ in the normalized coordinates. MCN uses cubic coefficient $M/6$. GRTR sets $\sigma_{\text{reg}} = \sqrt{\tilde{L}_2}/2$ and $r = 1/(4\sqrt{\tilde{L}_2})$. Its radius is $r\sqrt{\max\{\|g_{\text{raw}}\|, \epsilon\}}$; the Hessian shift is $\sigma_{\text{reg}}\sqrt{\|g_{\text{raw}}\|I}$. These practical tolerances do not carry the original oracle-error certificates. Each of 55 configurations has budgets $Q \leq 10^4$, $N_y \leq 10^6$, and 900 online seconds. Selection minimizes $(N_y^{\text{hit}}, Q^{\text{hit}})$ with fixed tie order, using only the true gradient target 10^{-6} . Native flags do not certify external success; independent evaluation and stagnation handling are specified in Section D.

The selected MCN/GRTR use 52.5%/70.0% fewer queries than fixed Minimax UTR-early (Table 4), and neither selected baseline rebounds above 10^{-6} after first hit. MCN nevertheless first raises P to 5.6712 (Figure 3). Of the practical grids, 20/30 MCN and 18/25 GRTR settings hit; all $\tau = 10^{-2}$ settings miss. For GRTR at $\tilde{L}_2 = .01$, tightening τ from 10^{-4} to 10^{-10} leaves $Q^{\text{hit}} = 229$ but raises queries from 1,073 to 10,523. More inner accuracy need not improve outer progress.

Formula-scale MCN/GRTR references with $M \approx 1.0373 \cdot 10^7$ and $L_2 \approx 2.0281 \cdot 10^6$ still have gradient norms .01485/.17086 after 10,000 subproblems. They also use different inner dynamics and formula-derived tolerances; this contrast is not solely an outer-scale effect. The supplied joint $\rho = 73.10$ was not independently globally certified ([Section D](#)).

Together, A demonstrates outer progress, B identifies inner-work mechanisms and their limits, and C shows the competitiveness of instance-tuned fixed-scale methods. We do not infer uniform empirical dominance, transfer of selected parameters to unseen starts, or improved test accuracy from these optimization experiments.

7 Conclusion

The parameter-free minimax method is a direct oracle realization of an inexact UTR theorem for general unconstrained optimization. The theorem separates the construction of approximate values and derivatives from outer progress; its supporting estimates provide the path and stored-value bounds needed for inner-work analysis. Corrected envelope oracles and persistent SCAR satisfy its hypotheses in the NC–SC setting, yielding leading complexities $O(\Delta_P \kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2})$ and $O(\Delta_P \kappa^2 \sqrt{\rho} \epsilon^{-3/2})$ for outer subproblems and inner-gradient evaluations, respectively.

The guarantee holds uniformly over each class with fixed common regularity constants once ϵ is sufficiently small. The algorithm does not certify that it has entered this regime. An observable all-accuracy certificate and a treatment of inexact trust-region solves remain beyond the present analysis.

A Proof of the inexact UTR theorem

All results in this appendix concern the general unconstrained objective F and the oracle interface of [Assumption 2.1](#). No minimax structure is used.

We first record the auxiliary estimates used by the minimax analysis. For this appendix, write $F^* = \inf_x F(x)$, $\Delta_F = F(x_0) - F^*$, $h_0 = \max\{\|\hat{g}_0\|, \epsilon\}$, and $\bar{\sigma} = \max\{\sigma_0, 2\sqrt{M/6}\}$, where $\hat{g}_0$ is the initialized working gradient. Define

$$B_\epsilon := \Delta_F + \frac{23C_v\epsilon^{3/2}}{7\sigma_0^3} + \frac{\epsilon^{3/2}}{512\sigma_0} \left[\left\lceil \log_2 \frac{h_0}{\epsilon} \right\rceil + 1 + \log_2 \left(1 + \frac{5C_g}{4\sigma_0^2} \right) \right]. \quad (\text{A.1})$$

Proposition A.1 (Counting and path estimates). *Under the hypotheses of [Theorem 2.1](#), every generated scale is at most $\bar{\sigma}$. If K, J, Q denote, respectively, the numbers of accepted trials, failed trials, and solved subproblems at termination, then*

$$K \leq 1024\bar{\sigma}B_\epsilon\epsilon^{-3/2}, \quad J \leq \left\lceil \log_2 \max \left\{ 1, \frac{\sqrt{M/6}}{\sigma_0} \right\} \right\rceil, \quad Q = K + J + 1. \quad (\text{A.2})$$

There are at most $J + 1$ validation calls. Every stored iterate and every finite prefix of accepted steps satisfy

$$F(x) - F^* \leq B_\epsilon, \quad \sum_{\text{accepted}} \sigma_k^2 \|d_k\|^3 \leq 8B_\epsilon. \quad (\text{A.3})$$

The bounds on K and J also hold for every finite prefix.

We prove the proposition together with [Theorem 2.1](#) at the end of this appendix.

A.1 A common trust-region estimate

Lemma A.1 (Common inexact trust-region step estimate). *Let $F : \mathbb{R}^n \rightarrow \mathbb{R}$ be twice continuously differentiable with an M -Lipschitz Hessian, $M > 0$. At a point x , let g and a symmetric H satisfy*

$$\|g - \nabla F(x)\| \leq \delta_0, \quad \|H - \nabla^2 F(x)\| \leq \xi.$$

For $a \geq 0$ and $R > 0$, let d be a global solution of

$$\min_{\|u\| \leq R} q(u) := g^T u + \frac{1}{2} u^T (H + aI) u,$$

let λ be the multiplier supplied by (2.1), and write $r = \|d\|$. Let a candidate gradient g^+ satisfy $\|g^+ - \nabla F(x + d)\| \leq \delta_+$. Then

$$F(x) - F(x + d) \geq \frac{a - \xi}{2} r^2 + \frac{\lambda}{2} r^2 - \frac{M}{6} r^3 - \delta_0 r, \quad (\text{A.4a})$$

$$\|g^+\| \leq (a + \xi + \lambda) r + \frac{M}{2} r^2 + \delta_0 + \delta_+. \quad (\text{A.4b})$$

Proof. Apply (2.1) with $B = H + aI$. Its stationarity and positive-semidefiniteness conditions give

$$q(d) = -\frac{1}{2} d^T (H + aI) d - \lambda r^2 \leq -\frac{\lambda}{2} r^2,$$

and Hessian Lipschitz continuity gives the standard Taylor estimates [22, Lemma 1.2.4]

$$\begin{aligned}\|\nabla F(x+d) - \nabla F(x) - \nabla^2 F(x)d\| &\leq \frac{M}{2}r^2, \\ F(x+d) - F(x) - \nabla F(x)^T d - \frac{1}{2}d^T \nabla^2 F(x)d &\leq \frac{M}{6}r^3.\end{aligned}$$

The second estimate and the current endpoint errors imply

$$F(x) - F(x+d) \geq -g^T d - \frac{1}{2}d^T H d - \delta_0 r - \frac{\xi}{2}r^2 - \frac{M}{6}r^3.$$

Since $-g^T d - d^T H d/2 = -q(d) + ar^2/2$, the model bound yields (A.4a).

For the candidate gradient, decompose the two endpoint errors, Taylor remainder, Hessian error, and model residual:

$$\begin{aligned}g^+ &= [g^+ - \nabla F(x+d)] + [\nabla F(x+d) - \nabla F(x) - \nabla^2 F(x)d] \\ &\quad + [\nabla F(x) - g] + [\nabla^2 F(x) - H]d + [g + Hd].\end{aligned}$$

The stationarity part of (2.1), again with $B = H + aI$, gives $g + Hd = -(a + \lambda)d$. Taking norms proves (A.4b). $\square$

A.2 Safe-scale acceptance and validation

Lemma A.2 (Safe scale for inexact UTR). *Under the hypotheses of Theorem 2.1, the following hold for every generated trial of Algorithm 1. If $\sigma^2 \geq M/6$, every ordinary trial of Algorithm 1 satisfies both acceptance tests, and every trial with $h = \epsilon$ and $r < R(h, \sigma)$ passes validation. At any scale $\sigma \geq \sigma_0$, successful validation implies*

$$\|\nabla F(x^+)\| < \epsilon, \quad \lambda_{\min}(\nabla^2 F(x^+)) \geq -\frac{85}{32}\sigma\sqrt{\epsilon}. \quad (\text{A.5})$$

Proof of Lemma A.2. Apply Lemma A.1 to F with

$$a = \sigma\sqrt{h} + \frac{\sigma\sqrt{\epsilon}}{64}, \quad \xi = \frac{\sigma\sqrt{\epsilon}}{64}, \quad \delta_0 = \delta_+ = \frac{\epsilon}{1024}.$$

The endpoint errors follow from (2.5). Since $r \leq \sqrt{h}/(4\sigma)$ and $M \leq 6\sigma^2$,

$$\frac{M}{6}r^3 \leq \frac{W}{4}.$$

Thus (A.4a) gives

$$F(x) - F(x+d) \geq \frac{W}{4} + \frac{U}{2} - \frac{\epsilon r}{1024}.$$

Young's inequality and $h \geq \epsilon$ imply

$$\frac{\epsilon r}{1024} \leq \frac{W}{8} + \frac{2\epsilon^2}{1024^2\sigma\sqrt{h}} \leq \frac{W}{8} + \frac{2T}{1024^2}.$$

After paying the two working-value errors,

$$\widehat{F} - \widehat{F}^+ \geq \frac{W}{8} + \frac{U}{2} - \left(\frac{2}{1024^2} + \frac{2}{4096}\right)T \geq \frac{W}{8} + \frac{U}{4} - \frac{T}{1024}.$$

For the gradient test, (A.4b) gives

$$\|\widehat{g}^+\| \leq \lambda r + \frac{h}{4} + \frac{h}{128} + \frac{3h}{16} + \frac{h}{512} = \lambda r + \frac{229h}{512} \leq \lambda r + \frac{h}{2}.$$

Hence every ordinary trial is accepted at a safe scale.

If $h = \epsilon$ and $r < R(h, \sigma)$, complementarity gives $\lambda = 0$, and the model shift is $a = (65/64)\sigma\sqrt{\epsilon}$. Applying (A.4b) to the true endpoint gradient ($\delta_+ = 0$), with $\delta_0 = \epsilon/1024$ and $\xi = \sigma\sqrt{\epsilon}/64$, gives

$$\|\nabla F(x+d)\| \leq \left(\frac{66}{256} + \frac{3}{16} + \frac{1}{1024}\right)\epsilon = \frac{457}{1024}\epsilon.$$

Moreover, model positive semidefiniteness, the current Hessian error, and Hessian Lipschitz continuity imply

$$\begin{aligned}\lambda_{\min}(\nabla^2 F(x+d)) &\geq -a - \xi - Mr \\ &\geq -\left(\frac{66}{64} + \frac{3}{2}\right)\sigma\sqrt{\epsilon} = -\frac{81}{32}\sigma\sqrt{\epsilon}.\end{aligned}$$

The validation error bounds therefore give

$$\|g_v\| \leq \frac{458}{1024}\epsilon, \quad \lambda_{\min}(H_v) \geq -\frac{82}{32}\sigma\sqrt{\epsilon},$$

so validation succeeds. At any generated scale, a successful validation and (2.6) imply

$$\|\nabla F(x^+)\| \leq \frac{\epsilon}{2} + \frac{\epsilon}{1024} < \epsilon, \quad \lambda_{\min}(\nabla^2 F(x^+)) \geq -\left(\frac{21}{8} + \frac{1}{32}\right)\sigma\sqrt{\epsilon}.$$

This proves (A.5). □

A.3 Accepted-step and scale-refresh budgets

For $h \geq \epsilon$, define the gradient level and stored potential

$$\mathcal{L}_\epsilon(h) := \lceil \log_2(h/\epsilon) \rceil, \quad \Phi := \widehat{F} + \frac{\epsilon^{3/2}}{512\sigma}\mathcal{L}_\epsilon(h). \quad (\text{A.6})$$

Lemma A.3 (Accepted-step potential decrease). *Every ordinary accepted step of Algorithm 1 satisfies*

$$\Phi - \Phi^+ \geq \frac{W}{32} + \frac{T}{1024} + \frac{U}{16}, \quad (\text{A.7})$$

where the scale is unchanged and Φ is defined by (A.6).

Proof. First note that if $h > \epsilon$ and $h^+ \leq \max\{h/2, \epsilon\}$, then

$$\mathcal{L}_\epsilon(h^+) \leq \mathcal{L}_\epsilon(h) - 1. \quad (\text{A.8})$$

For $\epsilon < h \leq 2\epsilon$, the two levels are 1 and 0. For $h > 2\epsilon$, the result follows from $\lceil t-1 \rceil = \lceil t \rceil - 1$. We will also use, for $c \geq 1$ and $h^+ \leq ch$,

$$\mathcal{L}_\epsilon(h^+) - \mathcal{L}_\epsilon(h) \leq \lceil \log_2 c \rceil \leq 1 + \log_2 c. \quad (\text{A.9})$$

Suppose first that the accepted step is interior. Then $\lambda = 0$ and $U = 0$. Since this is an ordinary trial, $h > \epsilon$. The gradient acceptance test implies $h^+ \leq \max\{h/2, \epsilon\}$, so (A.8) and the value test yield

$$\Phi - \Phi^+ \geq \frac{W}{8} - \frac{T}{1024} + \frac{T}{512} = \frac{W}{8} + \frac{T}{1024}.$$

This proves (A.7) in the interior case.

For a boundary step, put $z = \lambda r/h$. The gradient test gives $h^+/h \leq \max\{1, 1/2 + z\}$. By (A.9),

$$\mathcal{L}_\epsilon(h^+) - \mathcal{L}_\epsilon(h) \leq 1 + \log_2 \max\{1, 1/2 + z\} \leq 1 + \frac{z}{\ln 2}.$$

Since $r = \sqrt{h}/(4\sigma)$,

$$W = \frac{h^{3/2}}{16\sigma} \geq \frac{T}{16}, \quad U = \frac{zh^{3/2}}{4\sigma}, \quad Tz \leq 4U.$$

Consequently,

$$\begin{aligned} \Phi - \Phi^+ &\geq \frac{W}{8} + \frac{U}{4} - \frac{T}{1024} - \frac{T}{512} \left(1 + \frac{z}{\ln 2}\right) \\ &\geq \frac{W}{8} + \left(\frac{1}{4} - \frac{1}{128 \ln 2}\right) U - \frac{3T}{1024} \\ &\geq \frac{5W}{64} + \left(\frac{1}{4} - \frac{1}{128 \ln 2}\right) U. \end{aligned}$$

Finally, $5W/64 \geq W/32 + T/1024$ and $1/4 - 1/(128 \ln 2) \geq 1/16$, proving the claim. $\square$

To account for changes of scale, define

$$\begin{aligned} E(\sigma) &:= C_v \frac{\epsilon^{3/2}}{\sigma^3}, \quad E_0 := E(\sigma_0), \quad T_0 := \frac{\epsilon^{3/2}}{\sigma_0}, \\ \mathcal{L}_0 &:= \mathcal{L}_\epsilon(h_0), \quad D_0 := 1 + \log_2 \left(1 + \frac{5C_g}{4\sigma_0^2}\right). \end{aligned} \tag{A.10}$$

The budget in (A.1) can therefore be written as

$$B_\epsilon = \Delta_F + \frac{23}{7}E_0 + \frac{T_0}{512}(\mathcal{L}_0 + D_0).$$

Write $\sigma_* := \sqrt{M/6}$. Every failed trial doubles σ , and Lemma A.2 excludes failures at or above σ_* . If $\sigma_0 \geq \sigma_*$ there are no failures. Otherwise, at most $\lceil \log_2(\sigma_*/\sigma_0) \rceil$ doublings are possible. Consequently every finite prefix satisfies

$$J \leq \lceil \log_2 \max\{1, \sigma_*/\sigma_0\} \rceil, \quad \sigma \leq \bar{\sigma} := \max\{\sigma_0, 2\sigma_*\}. \tag{A.11}$$

Lemma A.4 (Cumulative accepted-step budget). *Under the preceding conditions, every finite prefix satisfies*

$$\sum_{\text{accepted}} \left(\frac{W_k}{32} + \frac{T_k}{1024} + \frac{U_k}{16} \right) \leq B_\epsilon. \tag{A.12}$$

Every stored iterate satisfies $F(x) - F^* \leq B_\epsilon$.

Proof. At a refresh, x is unchanged and the scale goes from σ to 2σ . By (2.4), the working-value jump is at most

$$E(\sigma) + E(2\sigma) = \frac{9}{8}E(\sigma).$$

The scales at failures form a doubling sequence, so the sum of all positive value jumps is at most

$$\frac{9}{8}E_0 \sum_{j \geq 0} 2^{-3j} = \frac{9}{7}E_0.$$

For the gradient-level term, the two working-gradient errors give

$$\|\hat{g}_{\text{new}} - \hat{g}_{\text{old}}\| \leq \frac{5C_g \epsilon}{4\sigma^2}.$$

The map $g \mapsto \max\{\|g\|, \epsilon\}$ is 1-Lipschitz and $h_{\text{old}} \geq \epsilon$. Hence

$$\frac{h_{\text{new}}}{h_{\text{old}}} \leq 1 + \frac{5C_g}{4\sigma^2}, \quad \mathcal{L}_\epsilon(h_{\text{new}}) - \mathcal{L}_\epsilon(h_{\text{old}}) \leq D_0.$$

Write $\mathcal{L}_{\text{old}}$ and $\mathcal{L}_{\text{new}}$ for these two levels. Since $T_{\text{new}} = T_{\text{old}}/2$ and the levels are nonnegative, the positive jump of the level contribution is at most

$$\left[\frac{T_{\text{old}}}{1024} (\mathcal{L}_{\text{new}} - 2\mathcal{L}_{\text{old}}) \right]_+ \leq \frac{T_{\text{old}} D_0}{1024}.$$

Its cumulative positive jump is therefore bounded by

$$\frac{T_0 D_0}{1024} \sum_{j \geq 0} 2^{-j} = \frac{T_0 D_0}{512}.$$

The initial potential is at most $F(x_0) + E_0 + T_0 \mathcal{L}_0/512$, and every stored potential is at least $F^* - E_0$. Telescoping (A.7) and paying the two refresh bounds gives

$$\sum_{\text{accepted}} \left(\frac{W}{32} + \frac{T}{1024} + \frac{U}{16} \right) \leq \Delta_F + 2E_0 + \frac{9}{7}E_0 + \frac{T_0}{512}(\mathcal{L}_0 + D_0) = B_\epsilon.$$

The same upper bound on the stored potential, together with $F(x) \leq \Phi + E_0$, gives $F(x) - F^* \leq B_\epsilon$. $\square$

A.4 Termination and the theorem's conclusions

Proof of Theorem 2.1 and proposition A.1. Every oracle call is finite by the interface assumptions. By Lemma A.2, failed trials can occur only below $\sqrt{M/6}$, and the scale bound (A.11) limits their number uniformly over all finite prefixes. Since $T_k \geq \epsilon^{3/2}/\bar{\sigma}$, Lemma A.4 gives, for every finite prefix,

$$K \frac{\epsilon^{3/2}}{1024\bar{\sigma}} \leq B_\epsilon.$$

Every nonterminal trial is either accepted or failed. Hence an infinite run would contradict one of these finite-prefix bounds, and the algorithm terminates.

At termination, there have been K accepted trials, J failed trials, and one successful validation trial, each solving one subproblem. Thus $Q = K + J + 1$, proving (A.2). Every validation either terminates or causes a failed trial, so there are at most $J + 1$ validation calls. The returned-point estimate (A.5) and $\sigma \leq \bar{\sigma}$ prove (2.11). For the cubic-path estimate, the radius constraint gives

$$4\sigma_k^2 r_k^3 \leq \sigma_k \sqrt{h_k} r_k^2 = W_k.$$

Thus Lemma A.4 implies $\sum_{\text{accepted}} \sigma_k^2 r_k^3 \leq 8B_\epsilon$ for every finite prefix. Its stored-value estimate supplies the other part of (A.3), proving Proposition A.1.

To obtain the total count (2.12), the initial working-gradient bound gives

$$\frac{h_0}{\epsilon} \leq \max \left\{ 1, \frac{\|\nabla F(x_0)\|}{\epsilon} + \frac{C_g}{\sigma_0^2} \right\}.$$

Consequently the bracket in (A.1) is at most a universal constant times $\log(e + \|\nabla F(x_0)\|/\epsilon + C_g/\sigma_0^2)$. Substitute this estimate and $\bar{\sigma} \leq \max\{\sigma_0, \sqrt{M}\}$ into (A.2). The terminal $+1$ is absorbed because this logarithm and $\max\{\sigma_0, \sqrt{M}\}/\sigma_0$ are both at least one, and the bound on J is $O(1 + \log(1 + \sqrt{M}/\sigma_0))$. This gives (2.12) and completes the theorem. $\square$

B Persistent SCAR

Derivation of Fact 2.1. For $u^* = \arg \min \phi$, the cited AR theorem with initial regularization $\nu/10$ gives

$$\|\nabla \phi(v)\| \leq \frac{\nu}{2} \|u - u^*\| \leq \frac{\nu}{2\mu} \|\nabla \phi(u)\|,$$

where the second inequality is strong convexity. This proves the halving implication. The theorem's gradient cost becomes $O(1 + \sqrt{L/\nu})$. At each AR stage the backtracking estimate is at most $\max\{\mathcal{M}_{s-1}/2, 2L\}$: a trial estimate at least L satisfies the tested smoothness inequality, and backtracking doubles its estimate only after a failed test. Starting from $\mathcal{M} \leq 4L$, induction gives $\mathcal{M}_s \leq 2L$ and hence the claimed bound on $\mathcal{M}^+$. The argument also covers $L = \mu$. $\square$

For completeness, the secant bounds (2.16) follow from strong monotonicity and Cauchy–Schwarz:

$$\mu \|z - u\|^2 \leq \langle \nabla \phi(z) - \nabla \phi(u), z - u \rangle \leq \|\nabla \phi(z) - \nabla \phi(u)\| \|z - u\|.$$

Dividing by $\|z - u\|^2$ gives the lower bound; the upper bound is gradient Lipschitz continuity.

Proof of Lemma 2.2. The initial line-search estimate satisfies $\mathcal{M}_0 \leq 4L$. Every AR call returns $\mathcal{M}^+ \leq 2L$ by Fact 2.1. All objectives have the same upper smoothness bound L , so this retained estimate remains admissible even when the next objective changes. Thus the AR guarantee applies inductively to every attempted call.

By Fact 2.1, a halving attempt on any ϕ_j must succeed whenever $\nu \leq \mu$. In particular, failure implies $\nu > \mu$ and the update after a failure leaves $\nu > \mu/4$. Since the initial guess satisfies $\nu_0 \geq \mu$, throughout the complete sequence we have $\nu \geq \mu/4$. After $\lceil \log_4(\nu_0/\mu) \rceil$ failures, the guess is at most μ , and no later objective can cause a failure: the same strong-convexity bound applies to every ϕ_j . Thus the complete sequence has at most $\lceil \log_4(\nu_0/\mu) \rceil$ failures, and every halving call terminates.

Each attempt uses $O(1 + \sqrt{L/\nu}) = O(\sqrt{L/\mu})$ gradient evaluations because $\nu \geq \mu/4$ and $L/\mu \geq 1$. There are at most N successful AR attempts and at most $\lceil \log_4(\nu_0/\mu) \rceil$ failed attempts. The at most N initial residual queries, including queries for calls with zero initial residual, fit in the same bound. Summing proves (2.17). Returning u when $\nabla \phi_j(u) = 0$ already satisfies the required residual reduction and does not change either retained estimate. $\square$

C Published-setting MCN reproduction

For reference, we retain the MCN experimental parameters $M = 10$, $K = 8$, $\eta = 0.1$, and $\theta = 0.9$ with the original 21-update budget. We use the paper’s Euclidean cubic model and the same global spectral solver as in [Section 6.1](#); this is not a claim to reproduce every implementation detail of the released coordinatewise cubic code. These runs are excluded from the tuned main comparison. Each uses 168 inner-gradient queries. The target in the table is exactly (6.1); a dash means no hit within 21 updates, not nonconvergence beyond that budget.

x_0	First hit T	Final $P - P^\star$	Final $\ \nabla P\ $
$(10^{-3}, 10^{-3}, 10^{-3})$	–	$5.74 \cdot 10^{-9}$	$4.79 \cdot 10^{-4}$
$(0, 0, 1)$	9	0	0
$(0.1, 0.1, 0.1)$	–	$5.67 \cdot 10^{-5}$	$4.76 \cdot 10^{-2}$
$(1, 0.1, 0.1)$	–	$6.64 \cdot 10^{-4}$	$1.10 \cdot 10^{-1}$

Reported zeros are floating-point values. Full trajectories are retained in the experiment’s published-setting reproduction record.

D Experimental protocols and reproducibility

Implementation and accounting. Minimax UTR and Minimax UTR-early use $c_0 = 1/128$ and $\sigma_0 = 1/\log(e/\epsilon)$ throughout. A charged secant in the all-ones unit direction initializes its curvature state without the strong-concavity modulus. The implemented model uses $h = \max\{\|\hat{g}\|, \epsilon\}$, radius $\sqrt{h}/(4\sigma)$, and Hessian shift $\sigma\sqrt{h} + \sigma\sqrt{\epsilon}/64$. Initial refinement targets $c_0 a_0 \sqrt{\epsilon}/(4\sigma_0)$, with observable secant a_0 ; subsequent calls follow the recorded tracking, refresh, and validation schedule. A uses `tracking_mode=target`. These operational settings are disclosed rather than identified with every subroutine setting of the analyzed algorithm. Paired ordinary/early runs differ only in the inner return test.

N_y counts performed inner-gradient evaluations, including checks, secants, and unsuccessful work. A’s fixed-count AGD uses K_{in} queries per call; C’s residual-stopped AGD uses $1 + 2k$ for k updates, plus the recorded initial query and interrupted work. A C query covers all 455 samples. C tables charge initialization and current-point model refinement; terminal points without a new model use arrival counts. Figures subtract initialization and place the common x_0 at zero. Schur assembly, Hessians, dense subproblem solves, and offline primal evaluation consume work outside N_y . C’s Minimax UTR value oracle uses an eight-point curvature integral for small-step remainders; it does not perform a hidden inner Newton solve. Offline roots never feed online updates. A uses analytic primal derivatives; B uses scalar bisection; C uses independent Newton solves in physical coordinates with each unaveraged sample residual at most 10^{-12} .

A: parameter searches. At each start, selection prioritizes the common stationarity hit, then $(Q^{\text{hit}}, N_y^{\text{hit}})$, then fixed grid order; invalid solves are ineligible. The following sets define the complete candidate families:

MCN:	$M \in \{.1, .3, 1, 2, 3, 5, 10, 30\},$	$K_{\text{in}} \in \{4, 8, 12, 20, 40\},$
	$\eta_y \in \{.05, .1, .15, .2\},$	$\theta_y \in \{0, .5, .8, 9/11, .9\};$
GRTR:	$K_{\text{in}} \in \{5, 8, 10, 20\},$	$\eta_y \in \{.05, .1, .2\},$
	$\theta_y \in \{0, .5, .9\},$	$(\sigma_{\text{reg}}, r) \in \{.03, .1, .3, 1\} \times \{.3, 1, 3\};$
HSDA:	$K_{\text{in}} \in \{4, 8, 12, 20, 40\},$	$\eta_y \in \{.05, .1, .15, .2\},$
	$\theta_y \in \{0, .5, .8, .9\},$	$\alpha_H \in \{.003, .01, .03, .1, .3\},$
	$\Lambda \in \{.03, .05, .1, .2, .3\},$	$\omega_H \in \{.05, .1, .25, .4\}.$

MCN exhausts 800 configurations per start. GRTR first tests 36 inner settings at $(\sigma_{\text{reg}}, r) = (.1, 1)$, then 12 outer settings, then one adjacent-coordinate pass, caching duplicates: 200 distinct configuration–start pairs overall. HSDA exhausts 8,000 settings per start. Baseline budgets are 100 updates; MCN/GRTR stop at external hit, while HSDA records its native flag and completes the short-step update without stopping. Its displayed curves end at offline first hit. Inner points are warm-started with momentum reset. The searches consume, respectively, 2,920,692, 132,327, and 53,760,000 queries, excluded from selected curves. Unequal, same-start searches are not held-out comparisons. The archive lists all selected parameters and complete trajectories.

B: bounds and pathwise verification. For the base payoff, bounds on the curved-valley Hessian and its Lipschitz constant are $L_0 = 3.03777417$ and $M_0 = 4.03998631$. The cosine term contributes at most 2η and $2\sqrt{2}\eta$, respectively. With $R = 6$ and $\eta = .1$, this yields

$$\ell = \max\{L_0, R\} + 2\eta = 6.2, \quad \mu = R - \eta = 5.9, \quad \rho = M_0 + 2\sqrt{2}\eta = 4.32282902.$$

The scalar envelope $\Psi(s) = \max_z \{\eta[1 - \cos(s + z)] - Rz^2/2\}$ satisfies $|\Psi'''| \leq \eta[R/(R - \eta)]^3$, giving $\bar{M}_P = 4.14515772$. The MCN formula scale 28.37671254 and $\bar{M}_P$ are chosen upper bounds, not measured optimal constants.

Replay freezes the 47 accepted-path points and incoming scales. The target is $r_t = c_0 \ell \sqrt{\epsilon} / (4\sigma_t)$: $1.79406573 \times 10^{-4}$ initially and $4.48516433 \times 10^{-5}$ thereafter. Costs restart at zero and exclude path-generation rejections and validation. Under $z = D_\gamma y$, every solver must satisfy $\|D_\gamma^{-1} \nabla_y f\| \leq r_t$: the primal and intrinsic accuracy are unchanged. AGD uses known $L_y = 6.1\gamma$ and $\mu_y = 5.9/\gamma$; SCAR retains or resets state as specified in the main comparison. At $\gamma = 1$, persistent/warm-restart/cold-restart totals 462/391/445 decompose into 47 residual checks each, 1/47/47 secants, and 414/297/351 halving queries. Downward- ν updates across $\gamma = 1, 2, 4, 8$ are $(0, 0, 1, 2)$, $(0, 0, 44, 89)$, and $(0, 0, 46, 92)$, respectively. These are observed state corrections, not identification of the true modulus or an additive causal attribution at identical inner points. All 940 returned outputs pass their common residual target. Exact-primal controls and full-SCAR replay are not a full minimax baseline leaderboard.

C: frozen inputs and references. The seed-2026 stratified split has 455 training samples (170 malignant, 285 benign) and 114 test samples. Standardization uses training data only. Reference centers use KMeans with `n_init=1` and seed 2026; amplitudes use $N(0, .1^2)$ from the same seed and the bias is zero. The frozen start adds a unit-scale seed-4108 Gaussian perturbation. It minimizes the initial primal Hessian eigenvalue over 33 predeclared candidates, without algorithm outcomes or test labels. The saved vector is shared by all methods. Geometry screening is separately charged.

The main text gives the 30 MCN and 25 GRTR practical grid points. They fix $(L_y, \mu_y) = (3.25, .175188)$ and tune an absolute residual tolerance, with no native stopping certificate. Analytic bounds give $L_y = 2 + 5/4$ and $\mu_y < 2 - C_y$, where $C_y = (5 + 25/(4e))/4$. The two formula-prescribed references instead use $(\ell, \mu, \rho) = (5.1283, .175188, 73.10)$ and AGD step/momentum $(.1949964, .6880101)$; the supplied ρ has not been independently certified here. Their outer scales are $M = 1.03728852 \times 10^7$ and $L_2 = 2.02809869 \times 10^6$, with residual targets 1.77922×10^{-10} and 3.55844×10^{-13} . Both miss 10^{-6} stationarity at 10,000 updates: final gradients are .01485070 and .17086028, using 318,891 and 474,269 queries. Selection uses initialization-inclusive $(N_y^{\text{hit}}, Q^{\text{hit}})$ and fixed grid order. Non-hits have no finite hitting cost; the best practical baselines use fewer queries than fixed Minimax UTR-early here.

C: stopping, truncation, and archived evidence. Budgets are 10,000 subproblems, 10^6 queries, and 900 online seconds. Only offline true gradient $\leq 10^{-6}$ defines success. An initial implementation ran to budget despite numerical stagnation; a later checkpoint-compatible patch stops after ten tiny-step/raw-gradient checks (or an unchanged iterate), labeling stagnation rather than success. Earlier completed records remain intact, so terminal costs mix numerical-stop versions; first-hit scores use the original histories. Display-only truncation removes a terminal sequence of at least 20 points if raw gradient $\leq 10^{-12}$ and step $\leq 10^{-10}$, or true gradient $\leq 10^{-10}$. It retains the first tail point and any first hit; earlier rebounds remain visible. This removes 474,015 display points from 50 curves without changing scores or actual costs.

All 57 server records were checked. Float64, single-threaded sequential runs used an Intel Xeon Platinum 8469C and PyTorch 2.8.0+cpu. Search costs were 1,817,112 queries, 27,957.3 online seconds, and 39,534.3 offline seconds, including idle tails. Minimax UTR-early ran on another machine, so no timing ratio is claimed. The accompanying experimental archive contains source hashes, exact manifests, complete grids, selected histories, state-event logs, scale diagnostics, and truncation audits; full server checkpoints are retained separately. These single-instance results establish neither held-out tuning performance nor uniform superiority.

References

- [1] Luo Luo, Yujun Li, and Cheng Chen. Finding second-order stationary points in nonconvex-strongly-concave minimax optimization. In *Advances in Neural Information Processing Systems*, volume 35,

pages 36667–36679, 2022.

- [2] Ziyi Chen, Zhengyang Hu, Qunwei Li, Zhe Wang, and Yi Zhou. A cubic regularization approach for finding local minimax points in nonconvex minimax optimization. *Transactions on Machine Learning Research*, 2023. URL <https://openreview.net/forum?id=jVMMdg31De>.
- [3] Tongliang Yao and Zi Xu. Two trust region type algorithms for solving nonconvex-strongly concave minimax problems. *Scientia Sinica Mathematica*, 56(2):437–454, 2026. doi: 10.1360/SCM-2024-0063.
- [4] Junlin Wang and Zi Xu. Gradient norm regularization second-order algorithms for solving nonconvex-strongly concave minimax problems. *Journal of Scientific Computing*, 105(2):41, 2025. doi: 10.1007/s10915-025-03069-8.
- [5] Jia-Hao Chen, Zi Xu, and Hui-Ling Zhang. A homogeneous second-order descent ascent algorithm for nonconvex-strongly-concave minimax problems. *arXiv preprint arXiv:2602.14058*, 2026.
- [6] Bohao Ma, Nachuan Xiao, and Junyu Zhang. Line-search and adaptive step sizes for nonconvex-strongly-concave minimax optimization. *arXiv preprint arXiv:2601.10086*, 2026.
- [7] Bohao Ma, Nachuan Xiao, and Junyu Zhang. A single-loop regularized Newton method for nonconvex-strongly-concave minimax optimization. *arXiv preprint arXiv:2606.07003*, 2026.
- [8] Yuntian Jiang, Chang He, Chuwen Zhang, Dongdong Ge, Bo Jiang, and Yinyu Ye. Beyond nonconvexity: A universal trust-region method with new analyses. *Journal of Scientific Computing*, 2026. arXiv:2311.11489.
- [9] Guanghui Lan, Yuyuan Ouyang, and Zhe Zhang. Optimal and parameter-free gradient minimization methods for convex and nonconvex optimization. *Mathematical Programming*, 2026. doi: 10.1007/s10107-026-02352-2.
- [10] Pierre Ablin, Gabriel Peyré, and Thomas Moreau. Super-efficiency of automatic differentiation for functions defined as a minimum. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 32–41. PMLR, 2020. URL <https://proceedings.mlr.press/v119/ablin20a.html>.
- [11] Guojun Zhang, Kaiwen Wu, Pascal Poupart, and Yaoliang Yu. Newton-type methods for minimax optimization. *arXiv preprint arXiv:2006.14592*, 2020. URL <https://arxiv.org/abs/2006.14592>.
- [12] Maher Nouiehed, Maziar Sanjabi, Tianjian Huang, Jason D. Lee, and Meisam Razaviyayn. Solving a class of non-convex min-max games using iterative first order methods. In *Advances in Neural Information Processing Systems*, volume 32, pages 14934–14942, 2019.
- [13] Tianyi Lin, Chi Jin, and Michael I. Jordan. On gradient descent ascent for nonconvex-concave minimax problems. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 6083–6093. PMLR, 2020.
- [14] Tianyi Lin, Chi Jin, and Michael I. Jordan. Near-optimal algorithms for minimax optimization. In *Proceedings of the Thirty-Third Conference on Learning Theory*, volume 125 of *Proceedings of Machine Learning Research*, pages 2738–2779. PMLR, 2020.
- [15] Siqi Zhang, Junchi Yang, Cristóbal Guzmán, Negar Kiyavash, and Niao He. The complexity of nonconvex-strongly-concave minimax optimization. In *Proceedings of the Thirty-Seventh Conference on Uncertainty in Artificial Intelligence*, volume 161 of *Proceedings of Machine Learning Research*, pages 482–492. PMLR, 2021.

- [16] Ruichen Jiang, Ali Kavis, Qiujiang Jin, Sujay Sanghavi, and Aryan Mokhtari. Adaptive and optimal second-order optimistic methods for minimax optimization. In *Advances in Neural Information Processing Systems*, volume 37, pages 94130–94162, 2024.
- [17] Ruichen Jiang and Aryan Mokhtari. Generalized optimistic methods for convex-concave saddle point problems. *SIAM Journal on Optimization*, 35(3):2066–2097, 2025. doi: 10.1137/24M1630475.
- [18] Jun-Lin Wang, Zi Xu, and Hui-Ling Zhang. A fully parameter-free second-order algorithm for convex-concave minimax problems. *Journal of Machine Learning Research*, 27(76):1–32, 2026.
- [19] David G. Luenberger and Yinyu Ye. *Linear and Nonlinear Programming*. Springer, Cham, 5 edition, 2021. doi: 10.1007/978-3-030-85450-8.
- [20] Stefania Bellavia, Gianmarco Gurioli, Benedetta Morini, and Philippe L. Toint. Adaptive regularization algorithms with inexact evaluations for nonconvex optimization. *SIAM Journal on Optimization*, 29(4): 2881–2915, 2019. doi: 10.1137/18M1226282.
- [21] Andrew R. Conn, Katya Scheinberg, and Luís N. Vicente. Global convergence of general derivative-free trust-region algorithms to first- and second-order critical points. *SIAM Journal on Optimization*, 20(1): 387–415, 2009. doi: 10.1137/060673424.
- [22] Yurii Nesterov. *Introductory Lectures on Convex Optimization: A Basic Course*, volume 87 of *Applied Optimization*. Kluwer Academic Publishers, Boston, 2004. doi: 10.1007/978-1-4419-8853-9.